Loki III: Reverse Engineering Structural Netlists without a Language Model
How far do provenance, a Bayesian L1 and a graph L2 get on their own?

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Abstract—Gate-Breaker showed that a language model can recover functional RTL from a gate-level netlist, and it ended by naming the graph techniques its model needed: clustering to name buses, a classification of signals as datapath or control, and partitioning. Loki III asks the complementary question: how much structure can be recovered without a language model, from provenance, a Bayesian hypothesis flow and graph analysis alone? This paper describes the benchmarks built to answer that question and reports the recovery statistics under the Gate-Breaker metrics. The benchmarks are 68 synthesizable micro designs and the 156 VerilogEval references (150 build on three floors), compiled by a provenance-tracked front end into named, labelled, anonymised netlist datasets; a renamed-instance protocol then proves the pipeline naming-neutral across 105,646 renamed identifiers. On the L1 hypothesis flow, the structures that are already reverse engineered by the synthesis flow recover almost perfectly (hard adders 67 of 67, hard multipliers 30 of 30), while the structures the flow erased recover weakly (macro-F1 0.354 on the first floor, 0.108 on the second, calibrated). A second round of compute-only experiments finds that re-mapping the gate netlist to LUT6 brings its in-sample recovery to the second floor's level (not on the held-out designs, where the result is inconclusive), that the language model is more readable than Loki on the same designs while Loki's output is formally equal to its input, and that an operator-recovery F1 sees what AST and CFG similarity cannot. A third round, also compute-only and with its decision rules fixed beforehand, finds that most wrong candidates of the uncalibrated detectors lie inside real constructs, but only by the letter of the rule: the result does not survive leaving out one design or the near-tautological enable fragments, and the fragments rarely tile the words. The L2 graph stage beats a 1-WL baseline on real micros once node, graphlet, WL and NPN features combine (macro-F1 0.4241 against 0.2717) and its clusters carry the designer's hierarchy (Louvain modularity 0.4220). We state the honest limits: no language model step exists, the recovered structure is structural rather than functional, and the paper's own determinism defects are reported.

Index Terms—reverse engineering, netlist, provenance, Bayesian inference, graph neural networks, FPGA CAD

I. Introduction

Gate-Breaker [1] takes a gate-level netlist with obfuscated wire names, formats it into a four-step prompt (interface, decomposition, function, generation), asks a language model to write RTL, and checks the result by simulation. On 156 VerilogEval problems [2] its best model recovered functional RTL for 75.1% of the runs, and 53.4% on the hardest quartile. The paper's own conclusion is the interesting part: external graph tools should help the model – clustering to name buses, a classification of signals as datapath or control, and partitioning of the netlist into smaller parts.

Loki III (https://github.com/drpaj12/loki) is the Yggdrasil III suite's reverse-engineering tool, and it targets exactly those three items – with no language model step at all. Its pipeline is L0, a provenance ground-truth dataset made from the benchmarks; L1, a Bayesian hypothesis flow whose pattern detectors propose structures and whose proofs confirm them; and L2, graph analysis (clustering, structural features, graph neural networks) over the same netlists. The research question this paper answers is: how good are the L0, L1 and L2 experiments, without more intelligence, at recovering structure? The answer matters for two audiences. For the Gate-Breaker line, it prices the graph side of the future work the paper names. For reverse engineering, it measures what a provenance-tracked front end and honest statistical machinery can do before any model is added.

The contributions are: (1) a benchmark method – provenance-tracked, labelled, anonymised netlist datasets with a renamed-instance protocol that proves naming neutrality, so the pipeline can be evaluated on designs whose names carry nothing; (2) the L1 and L2 recovery statistics under the Gate-Breaker metrics and under Loki's own (precision, recall, calibration, clustering quality, GNN against a 1-WL baseline); (3) the honest limits, including two determinism defects the experiments found.

II. Background

A. The provenance problem and Odin III

Synthesis modifies the designer's functional intention: flattening removes hierarchy, optimization merges oper-

ators, and mapping covers logic with cells that carry no source names. Linking each netlist object back to the HDL that produced it is the provenance problem [3]. Odin III, the provenance-tracked front end under Loki, keeps that link: every object of its one IR records its source location, and the link survives elaboration, optimization, partial mapping and the linked ABC flow. Loki stands on that link: its L0 datasets carry the answer key the synthesis tool kept, which is what makes the recovery statistics in this paper possible at all.

B. Gate-Breaker as the method baseline

Gate-Breaker measures recovery by functional correctness and by the similarity of the recovered and reference RTL: Success Rate (the fraction of K samples that simulate correctly), Success@k (the fraction of designs solved within the first k samples), AST similarity (mean subtree Jaccard over normalized identifiers, depth $d = 3$) and CFG similarity (a normalized shortest-path node kernel over control-flow graphs). Its complexity score weights sequential blocks, generates, loops, instantiations, always blocks, buses, multiplexers, cases, arithmetic operators, assignments and concatenations into quartiles Q1 (simple combinational) through Q4 (complex sequential). Loki reports its own statistics on the same axes where they apply – recovery rate per structure kind, calibration error, clustering quality against the known hierarchy, and GNN accuracy against a 1-WL baseline – and the Gate-Breaker metrics on the replication runs.

C. Dataflow analysis and HAL as a second baseline

HAL, the Hardware Analyzer of the Embedded Security Group [4], [5], is an open framework (MIT licence) for the reverse engineering of gate-level netlists. Its dataflow plugin implements DANA [6], which groups flip-flops into register words from their shared control nets (clock, enable, set, reset) and from common predecessors and successors in the flip-flop dataflow graph, without reading the logic between them. DANA is therefore a word-finding method that is independent of the cell library once the flip-flops are known, which makes it the natural external baseline for the register words that Loki's word-first detector (Section V-I) targets. Section V-J runs both on the same netlists and scores both against the same Odin III answer key. We credit HAL and DANA in the README and the bibliography; no HAL code is part of this repository, only our gate-library files, a converter and the scorer.

III. Methodology: floors, pipeline and evaluation

A. Floors: the abstraction level of the netlist

A floor is the abstraction level and technology mapping at which an Odin III design is cut and written out as the L0 netlist, the object the reconstruction receives. Every floor starts from the same elaborated design, the common front proc; fsm; opt; flatten; opt -full; wreduce (FRONT in loki/odin.py), and then maps it with a floor-specific script (floor_script in the same file). Table I lists the floors. Four exist: f1-k6 and f1-earch (one floor per FPGA architecture file; the directory name is f1- plus the architecture's lower-case prefix), f2, and f3.

What a floor's netlist contains decides what is left to reconstruct. At f1 the flow's hardmap step has already recognised adders, multipliers and memories, so they appear as typed hard cells; the construct they implement is part of the netlist, and the labels still name it. At f2 the same logic is only LUT truth tables: an adder is a carry chain of LUTs that has to be found again. At f3 the logic is a network of two-input gates, the form Gate-Breaker's language model receives. The f2 output does not depend on the architecture file, and f3 reads none, so their files are byte-identical across architectures.

Loki uses several floors because what a synthesis flow erases depends on the mapping, and recovery has to be measured against each. f1 keeps the structures the flow already reverse engineered, which is what lets the hard detector of L1 read them from Odin III's unmap facts; f2 removes that crutch with the same source and the same answer key; f3 makes the input comparable to Gate-Breaker's and is the floor on which the AST and CFG similarities are computed (Section V-C). Within one design the floors share one source of truth, the provenance spans of the same elaborated IR, so a difference between floors is a difference of the mapping.

B. L0: provenance ground-truth datasets

The first benchmark set is the 68-design micro suite (counters, registers, arithmetic, ALUs, muxes, comparators, small FSMs) that Odin III already compiles and checks against Yosys; the second, the 156 VerilogEval references, is described in Section IV. L0 compiles each design on every floor of Table I. Every netlist object carries its provenance (source file and line), a neutral nk name from one shared counter, and a label: the structure kind (construct_type) is the answer key in labels.csv, and the netlist-visible hard-block kind (cell_type) travels in the graph labels, so structures that are already reverse engineered stay visible to the reconstruction. loki check validates every dataset: 68 designs, 269 floors, a label row for every node, zero errors. The two hard-block designs the v1.0 front end refused now compile on the v1.1.0 SystemVerilog front end, so the set is complete.

Renamed instances. A separate protocol asks the naming question directly: rename every identifier of every netlist (105,646 identifiers across 269 netlists) through a shared permutation, run the pipeline on the renamed floors, and inverse-map every token back. The verifier reports token-for-token equality with the original, zero failures, and the base and renamed score digests are byte-identical (sha256 over all run scores, 204 of 204 pairs) – the pipeline is naming-neutral, so nothing it recovers is an artefact of a name.

TABLE I
The floors of L0. All floors run the common front first; the mapping script is the second part of floor_script. Node kinds are those of graph.json (pi, po, const, lut, ff, hard, gate).

Floor	Reads	Mapping script	Netlist contents
f1-k6	read_arch k6_frac_N10_ frac_chain_mem32K_40nm	hardmap; lower; abc –lut 6	LUT6, flip-flops, and hard adder, multiplier and RAM cells that keep their Odin III type
f1-earch	read_arch of the EArch file	same as f1-k6	same kinds, EArch hard blocks
f2	read_arch (cell types only)	lower; abc –lut 6	LUT6 and flip-flops only; arithmetic is soft, so no hard node exists
f3	read_techlib lib/cmos6.o3lib	lower; abc –gates cmos6; dfflibmap cmos6	library gates (BUF, INV, NAND2, NOR2) and DFF flip-flops

C. L1: the Bayesian hypothesis flow

L1 is a round-based hypothesis flow over one netlist, with no language model. It reads a floor netlist into a fresh Odin III design (level 0, never edited) in which every cell is a function, and keeps a working copy that grows one level of hierarchy per committing round.

Hypotheses and detectors. A hypothesis is a claim that a set of level-0 cells is a structure of some kind, for example: a flip-flop with enable or synchronous reset (ff_en, ff_srst), a register, a bus of isomorphic slices, an adder, subtractor, comparator or constant comparison (eq_const), a mux or mux_tree, a counter, shift_reg, lfsr or fsm, a datapath or control unit, a per-cell role, or a hard block (hard_add, hard_mul, hard_mem). Nine detector plugins (arith, bus, ff_pattern, fsm, hard, mux, role, seqfn, unit) propose them. A detector reports features as bin labels (a width of 5 to 8, whether an enable function is exact, the NPN class of a cell) and never a likelihood; the likelihoods come from calibration.

Bayesian inference. Each hypothesis h of kind k from detector d carries a log-odds $L(h)$, the logarithm of the odds that it is a real structure, and $p(h) = 1/(1+e^{-L(h)})$ is its probability. L is a naive-Bayes sum: a prior plus evidence items, each the log-likelihood ratio (LLR) of an observation,

$$L(h) = \pi_{d,k} + \mathrm{clip}_{\pm 6.91}\Big(\sum_f w_{\phi(f)}\, \lambda_{k,f}(b_f) + \min\big(1.10\, n_{\mathrm{conf}},\, 3.30\big) - 2.30\, n_{\mathrm{ref}} \Big) + e_h, \quad (1)$$

where $\pi_{d,k}$ is the prior log-odds of that detector and kind; $\lambda_{k,f}(b_f)$ is the LLR of feature f in bin b_f; $w_{\phi(f)}$ weights the feature's family (structural, WL colour, NPN class or L2; the f3 calibration uses the first three only); n_{conf} and n_{ref} are the numbers of confirmed (or committed) and refuted child hypotheses (a bus slice, a register's flip-flops), which raise and lower their parent; and e_h collects fixed records (a sampled check adds $\ln 100 = 4.61$). The clip at $\pm \ln 1000 = 6.91$ limits the double counting that a naive-Bayes sum of correlated features would otherwise allow. A proof sets $L(h) \geq \ln 9999 = 9.21$ (the odds become 9999 to 1), and a counterexample sets $L(h) = -\infty$.

All three tables are calibrated from the training split of L0 (loki calibrate). The flow runs with default tables on every training design; each hypothesis is labelled true when it matches an L0 construct (Section III-E). The prior is the Laplace-smoothed precision of the detector, $\pi_{d,k} = \mathrm{logit}\,\frac{tp+1}{n+2}$ over n proposals of which tp are true. For example, the f3 comparator detector made 31,036 proposals of which 2 were true, so $\pi = \ln \frac{3}{31035} = -9.24$; the committed value is -9.244259 (experiments/f3-recovery/calib/f3.json). A bin's LLR is $\lambda = \ln \frac{(pos+1)/(POS+B)}{(neg+1)/(NEG+B)}$, with pos and neg the true and false proposals in the bin, POS and NEG the totals of the kind, and B the number of bins. The family weights w are the grid values in $\{0, 0.1, \ldots, 1\}$ that maximise the Bernoulli log-likelihood of the labels, fitted by three sweeps of coordinate ascent on out-of-fold rows (two folds, split by design). Without calibration (–calib none) the priors are 0 and the LLRs are the defaults of bayes.py. The reliability check is the expected calibration error over ten equal-width bins of the predicted probability, $\mathrm{ECE} = \sum_b \frac{n_b}{N} |\bar{p}_b - r_b|$ with r_b the observed match rate.

Proof engines. A hypothesis with a functional claim ("these 9 LUTs compute $A + B$ at width 5") can be proved. Three engines prove claims: cone_check first when the claim has a template, the truth table otherwise or when the cone cannot be bound to the template, and transfer after an exact proof. (i) cone_check, an Odin III pass, proves that the claim's cone of level-0 cells is equivalent to a template word cell (for example $add, $sub, $eq, $mux) with its ports bound to the cone's boundary nets, under a timeout; the reference functions of the claims (claims.py) cover addition, subtraction, the six comparisons, constant comparison, 2:1 and N:1 multiplexers, counters, shift registers and LFSRs. (ii) The truth-table prover enumerates the claim's inputs and side nets when together they number at most 16, an exact proof or a counterexample, and otherwise simulates 4096 seeded random vectors, which can refute a claim but only raises it by $\ln 100$. (iii) Proof transfer groups open hypotheses by operation, parameters and the 1-WL colour of their output drivers; when one member is proved exactly, the others are tried by a lockstep cone_match of the two cones, and one transferred member per group is proved for real as a spot check, revoking the transfers of the group if it fails. A sequential claim is proved as a next-state function (state

in, next state out); an FSM gets no proof.

Deciding the structure. Hypotheses that claim the same cells compete. Two hypotheses conflict when both kinds are exclusive, their cell sets intersect, and neither is a strict subset of the other (a strict subset is nesting: the smaller becomes a child of the larger). For each connected component C of the conflict graph the competitive probability is

$$p_{\text{comp}}(h) = \frac{e^{L(h)}}{1 + \sum_{j \in C} e^{L(j)}}, \tag{2}$$

where the "1" is the option that none of them is real; a hypothesis without conflicts has $p_{\text{comp}} = p$. Two confirmed hypotheses with equal cells are two true forms of the same logic and count as one member of C; the one whose kind ranks higher (a counter above an adder, for example) wins at commit and the other is kept as an alternative. A proof is spent on hypotheses with $p_{\text{comp}} \geq 0.2$, most probable first, at most 200 per round; a hypothesis below 0.01 is rejected, and re-enters when later evidence lifts it. A hypothesis is selected for commit when $p_{\text{comp}} \geq 0.9$, every child that could commit is already committed, and it does not conflict with a committed piece. Selection is greedy in the order (p_{comp} descending, kind rank, cell count descending, id), and the losers of a conflict become rejected or alternative.

Commit and the final structure. A selected hypothesis is rebuilt on a copy of the working design: a hypothesis is raised to a single word cell only if its kind is one of adder, subtractor, comparator or multiplexer, its claim has a cone_check template, and it is proved by cone_check (directly or by transfer); the cone is re-checked by cone_check unconditionally before the raise. A claim proved only by truth table is wrapped, not raised, and counters, shift registers, LFSRs, FSMs, registers, constant comparisons, multiplexer trees and carry-in adders are never raised. Every selected piece except buses and roles (which only set attributes) is wrapped by submod in a module named for its kind, whose instance is tagged with the hypothesis id, kind, confidence p_{comp}, proof status and the recovered function. If raise or submod fails, the offending hypothesis is rejected with a record of the error and the commit is retried once without it (reject_failed). The structure that L1 reports is therefore exactly the set of committed pieces in design.v; the alternatives, rejected and open hypotheses stay in overlay.json with their evidence. Algorithm 1 summarises the loop. It stops when a round commits nothing, proposes nothing new and changes no log-odds by more than 10^{-6} (a fixed point), after eight rounds, or at the time budget, then flattens and lowers the final design and checks it against the input with equiv-check; a result of differs is a bug and the run exits with an error.

The hard detector reads the unmap pass's facts: a hard adder, multiplier or memory that the flow already reverse engineered is recovered directly, with the decline reasons the pass recorded (carry-in chains and the like) kept as open hypotheses. The hard detector does not run on f2 or f3.

Algorithm 1 The L1 round loop (loki/re/loop.py).

```
level_0 ← read netlist as functions; work ← level_0
if floor f1 with hard cells then run unmap on work
for r = 1 to 8 do
    propose: every detector adds hypotheses over the view of work
    score: L(h) by Eq. 1; compete: p_comp by Eq. 2
    prove claims with p_comp ≥ 0.2: cone_check, truth table, transfer
    re-score and compete; reject p_comp < 0.01
    select p_comp ≥ 0.9 (children first); commit by raise and submod
    if nothing committed, nothing new, |ΔL| ≤ 10^−6 then
        break
    end if
end for
write design.v, overlay.json, report; equiv-check against the input
```

D. L2: graph analysis

L2 reads the L0 graph.json files and nothing else: never key.json and never the BLIF. It has three parts.

Structural features. Each node gets numeric features (fan-in, fan-out, logic level, LUT width, hard-port count), its kind and, on hard cells, its cell_type, plus three families: colours of 1-WL colour refinement (a hash of the node's label and the multiset of its neighbours' colours, per round, so that equal colours mean equal unfolded neighbourhoods), the NPN class of a LUT's function (invariant under permutation and negation of the inputs and the output), and the ORCA orbit counts of the 4-node graphlets around the node. Before graphlets are counted, the global signals (clock, reset, constants, global enables) are classified by role and their edges removed, so that a clock does not connect everything.

Clustering. Four methods partition a netlist graph into candidate blocks: k-means on the feature space ($k = \text{clip}(\text{round}\sqrt{N/2}, 2, 200)$), affinity propagation on a sample of at most 3000 nodes with the remaining nodes assigned to the nearest exemplar, spectral clustering (a spectral embedding of the largest connected component, then k-means), and Louvain community detection (resolution 1, no k). The first two use node features; the last two use the graph structure. A partition is scored by its modularity on the undirected simple projection of the graph, by its NMI against the known module_path and construct_type labels and by its ARI against module_path only over the LUT, flip-flop, hard and gate nodes. These two numbers are a quality report, never an input to a method: modularity says whether the clusters are dense subgraphs, NMI whether they are the designer's modules.

GNN against 1-WL. A node classifier predicts each node's construct_type: a two-layer GraphSAGE (GCN as the variant) of hidden width 64 with ReLU and dropout

0.2 over the graph and its reverse edges, trained with Adam (learning rate 0.01, weight decay $5 \cdot 10^{-4}$) for a fixed 100 epochs, full-batch, with class weights $1/\sqrt{\text{count}}$. The split is L0's, by design, with the class vocabulary taken from the training designs only (rarer values, and values first seen in a test design, are one class ___other___); there is no validation set and no early stopping, so nothing is tuned on a test design. Loss and metrics cover only the LUT, flip-flop, hard and gate nodes with a known target. The question is whether message passing earns anything over what the colour refinement already knows, because a message-passing GNN is at most as discriminating as 1-WL [7]. So the 1-WL baseline predicts, for each node, the most common training class of its WL colour (backing off from round 4 to round 0 when a colour has fewer than five training nodes), and the GNN has beaten it only if its test macro-F1 is higher. Two further baselines are reported on the same nodes, the majority class and a logistic regression on the features alone. The task is that of GNN-RE [8], which maps each gate of a flat netlist to the module it belongs to; Loki's model is a baseline for the same task, not a re-implementation, and the numbers are not compared with that paper's.

E. Evaluation protocol: how L1 and L2 are scored

Gate-Breaker evaluates a stochastic model: for each design the model is sampled K times, each sample's RTL is extracted and simulated against the VerilogEval testbench, and the results are aggregated per complexity quartile as the success rate (the fraction of the K samples that pass), Success@k (the fraction of designs with a passing sample among the first k), and the AST and CFG similarity of the recovered RTL to the reference. Loki's L1 and L2 are deterministic and their output is a structure, not an RTL text, so the same four questions are asked in different terms. Table II sets the two side by side.

L1, step by step.

1) Split. L0 assigns each design to train or test by a hash of the design name (sha256(seed:design) mod $100 < 80$ is train), so the split is fixed by the name and independent of the run.
2) Calibrate on the training designs of one floor (loki calibrate): the priors, bin LLRs, family weights and reliability curve of Section III-C.
3) Run loki re on every test netlist, calibrated and, as a diagnostic, with –calib none. The run never reads the answer key.
4) Ground truth. From words.json, the L0 word cells give ground-truth instances (type, F_g) with F_g the set of final cells of the word; counters, mux trees and FSMs are derived from the word connectivity.
5) Count and match. A hypothesis counts when it is committed or its $p_{\text{comp}} \geq 0.5$ (–score-p); it matches a ground-truth instance of an accepted type when the Jaccard of their cell sets is at least 0.5, greedily one-to-one by Jaccard. Precision is matched over counted, recall matched over ground truth, and average precision sweeps the threshold from 0.1 to 0.9.
6) Buses, roles and hierarchy. A bus matches a ground-truth bus (nodes sharing a module path and bus name, at least two bits) at the same Jaccard; its bit order is scored exactly and up to reversal. Roles are scored as precision and recall of the predicted control and datapath cells. The hierarchy similarity compares the clusters of the module paths to the committed pieces with the Dice score $F(g,h) = 2|g \cap h|/(|g| + |h|)$, $\text{HS}_{gt} = \sum_g |g| \max_h F(g,h) / \sum_g |g|$ and its converse HS_{l1}, and reports their harmonic mean.
7) Aggregate. loki score writes one score.json per run and loki score-all the floor-by-kind tables, the reliability curve of p_{comp} with its ECE, and, with –compare, the ablation against –features none.
8) Naming neutrality. The renamed twin of the dataset is run and scored against the original-named key; the base and renamed score.json must be byte-identical.

Of the Gate-Breaker metrics, the AST and CFG similarities are used on f3 as in the paper: AST similarity is the mean, over the subtrees of the reference, of the best Jaccard similarity to a subtree of the candidate (depth 3, node kinds normalised); Gate-Breaker computes both on functionally correct samples only, Loki's are unconditional; CFG similarity is the node kernel over shortest-path histograms of the control-flow graphs. They are computed twice per design, for the gate netlist (L0-stop) and for design.v (L1-stop), and the L1 delta is their difference. The success rate and Success@k need samples; a deterministic flow has one attempt per design, so they are reported only for the language-model replication (Section V-E). Formal final_equiv does not play the role of the testbench: it compares the flattened final design with the input netlist, so it shows that the rewrite preserved the netlist (it is trivial on f3, where L1 commits nothing), not that the design is correct against the reference RTL. Loki has no behavioural pass rate against the reference.

L2, step by step. (1) The same split applies. (2) loki analyze cluster writes a partition per method and design, with its modularity and NMI/ARI against module_path and construct_type; the table reports the mean over designs. (3) loki analyze gnn fits the vocabulary and the model on the training designs and writes metrics.json: accuracy and macro-F1 on the training and test nodes for the GNN, the majority class, the feature-only logistic regression and the 1-WL baseline, and beats_wl. (4) The neutrality check re-runs both on a permuted copy of every identifier.

F. One design through the pipeline

The steps below follow one design, the 4-bit comparator micro bm_DL_4_bit_comparator (three comparison

TABLE II
The Gate-Breaker evaluation and the Loki evaluation, step by step.

Step	Gate-Breaker	Loki
Input	cmos6 gate netlist of a VerilogEval reference, internal nets renamed n_⟨SHA-1 prefix⟩, ports kept	a floor netlist (f1, f2, f3) with names nk; for the neutrality check a renamed twin
Producer	a language model, K samples per design	the L1 flow, one deterministic run per design (fixed seed); no sampling
Correctness	simulate the extracted RTL against the testbench: success rate, Success@k	equiv-check of the flattened final design against the input netlist: final_equiv (formal; on f1 hard cells are cut points and only soft logic is checked). A self-consistency check of the rewrite, not correctness against the reference RTL; Loki has no behavioural pass rate against the reference
Match to the truth	AST and CFG similarity to the reference RTL	each counted hypothesis against the L0 answer key: Jaccard ≥ 0.5, one-to-one; precision, recall, F1, average precision per kind and floor; hierarchy similarity; AST and CFG of design.v against the reference on f3, unconditionally (Gate-Breaker's are computed on functionally correct samples only)
Aggregate	per complexity quartile	micro-averaged per kind and floor over the test split; AST and CFG per quartile of f3 node count

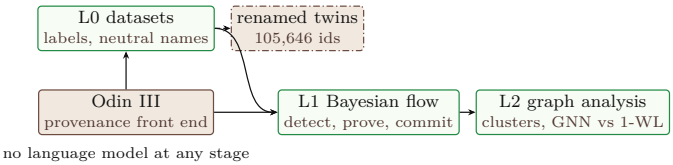


Fig. 1. The Loki III pipeline. L0 compiles the benchmarks through Odin III into provenance-tracked, labelled, anonymised datasets (and their renamed twins). L1 proposes and proves structures in a Bayesian flow that commits into the same IR. L2 reads the graphs back and scores clustering and a GNN against a 1-WL baseline. No stage has a language model.

outputs), from source to score. Each step names the file or code that defines it.

1) HDL to Odin III with provenance. read_verilog and the common front elaborate the design. Every IR object keeps its source spans (Node.sources(), the only provenance the binding exposes). Before mapping, Loki snapshots every word-level cell (type, parameters, spans) and every instance site.
2) Map to the floor. lower keeps each word cell's exact span on every gate it creates; abc –lut 6 gives each LUT the union of the spans of its cone. A LUT that merges the three comparators therefore lists all three spans.
3) Join and label. A final cell is matched to a word cell when the word's span set is a subset of the cell's. The primary match gives construct_type (here $eq, $gt, …); a cell with several matches is flagged (n_construct_cells above one) and a cell with none stays unknown: a label is an answer key, never a guess. The module path comes from the instance spans, bus name and bit index from net names and aliases (AeqB[0]), and the datapath or control role from a rule table.
4) Anonymise and write the L0 files. Nets, cells and port bits get names nk from one counter. Loki writes netlist.blif and netlist.v, graph.json (nodes with features and labels, edges), labels.csv (one row per node), words.json (the word-level connectivity the scorer needs) and key.json. Only key.json holds original names; no analysis reads it.
5) Verify. Odin III's equiv-check compares the written BLIF with the floor's own netlist and the anonymised netlist with the original; both must read equivalent before the floor counts as built.
6) L1: normalise. loki re reads netlist.blif (never key.json) into a fresh Odin III design, level 0, in which every logic cell is a function (a LUT by its truth table, a gate by its library function). On f1 with hard cells the unmap pass runs on the working copy and its facts feed the hard detector.
7) L1: hypothesis rounds. Each round runs the detectors, scores every hypothesis (Section III-C), proves the ones worth a proof, resolves conflicts and commits the confident ones (Algorithm 1); the next round sees the committed level.
8) L1: commit and output. A committed hypothesis is rebuilt as a word cell (raise) wrapped in a module (submod) tagged with its kind, confidence and recovered function. At the fixed point Loki writes design.v, overlay.json (every hypothesis with its evidence) and report.json, and checks the final design against the input with equiv-check.
9) L2. loki analyze reads graph.json only: structural features, four clusterings and a GNN (Section III-D).
10) Score. loki score compares the overlay with the answer key (Section III-E); loki score-all aggregates every run into per-kind tables.

In this micro the f3 comparator, as re-synthesised by the minimal .v front, has 26 logic cells (L1 spec, Section 3); the f2 LUTs carry the comparator spans that the join reads back.

IV. The VerilogEval benchmark and its structure

A. VerilogEval spec-to-rtl as the Gate-Breaker benchmark

Gate-Breaker evaluates on VerilogEval [2], taking the spec-to-rtl variant: 156 problems, each a natural-language prompt, a reference solution (*_ref.sv, module RefModule) and a testbench (*_test.sv). The Gate-Breaker flow synthesizes the reference to a cmos6 gate netlist, hands the netlist to the model, and checks the model's RTL against the testbench. Loki uses the same 156 references as L0 sources and reads the reference module as the design; the prompts and testbenches are not inputs to L0, L1 or L2. The references are SystemVerilog, read through Odin III's slang adapter (Section V-A).

B. The complexity score and its quartiles

Gate-Breaker groups the problems by a complexity score (its Eq. 2), a weighted sum of counts of HDL constructs:

$$C = 1000\,c_{\mathrm{Seq}} + 200\,(c_{\mathrm{FG}} + c_{\mathrm{F}}) + 20\,c_{\mathrm{Mo}} + 15\,(c_{@A} + c_{\mathrm{B}}) + 10\,(c_{\mathrm{M}} + c_{\mathrm{Ca}}) + 5\,c_{\mathrm{E}} + 3\,c_{\mathrm{A}} + 2\,c_{\mathrm{C}}. \quad (3)$$

The paper names the terms but does not define them, so the implementation we reuse (complexity.py of the second Gate-Breaker replication) fixes the reading: c_{Seq} counts clocked always blocks (a posedge or negedge in the sensitivity list), c_{FG} for-generate constructs, c_{F} for loops, c_{Mo} module instantiations, $c_{@A}$ all always blocks, c_{B} declarations wider than one bit, c_{M} multiplexers (ternary operators plus non-default case items), c_{Ca} case statements, c_{E} arithmetic operators (+ - * / %), c_{A} continuous assignments and c_{C} concatenations. A clocked block weighs 1000 on its own, so a design in which the counter recognises one scores at least 1000, which is Q3 or Q4.

The quartiles are the paper's absolute bins (its Table II), not rank quartiles of the benchmark: Q1 is 3 to 23, Q2 24 to 111, Q3 112 to 1064 and Q4 1065 to 1326. The 156 references fall 38, 39, 40 and 39 into Q1 to Q4, close to the paper's 39 per quartile. The 150 designs that build on every floor are split 38, 39, 37 and 36: the six missing ones are Prob031, Prob046 and Prob145 (Q3) and Prob078, Prob151 and Prob156 (Q4). Table III gives the counts.

Deviation: pyverilog AST against a regex fallback. The score is computed on the pyverilog AST of the first module when the file parses, and by regular-expression counts of the same terms otherwise, with the method recorded per design. Because the references are SystemVerilog, pyverilog parses only 60 of the 156; the other 96 use the regex counts. The split is skewed by quartile (Table III): 36 of Q1 and 23 of Q2 are AST-scored, but 1 of Q3 and none of Q4. The regex counts operator characters and keywords textually, so they differ from an AST count (the observed Q4 range, 1069 to 3259, exceeds the paper's bin top of 1326 for this reason or because the problems differ; we do not separate the two). The regex also never counts always_comb, always_ff or always_latch blocks, which 34 references use, so c@A and cSeq are undercounted in the 96 regex-scored designs, most of Q3 and Q4; the quartile split and the feature trends below rest on approximate scores. The 156 scores equal those of the replication that computed them first, exactly, so the quartiles here and in Table XII are the same partition.

C. The feature matrix

Gate-Breaker characterises its benchmark by the HDL structures it contains. We build the same kind of matrix for the 150 built designs, from the f3 floor of each, with rules fixed in advance (summarize.py): arithmetic counts the word cells of type $add $sub $mul $div $mod $neg $pos $pow $alu $macc; select/mux those of type $mux $pmux $bmux $demux; storage the register words ($dff, $dffe, $adff, $adffe, $sdff, $sdffe, $sdffce, $dffsr, $aldff, $dlatch, $adlatch); FSM is 1 when Odin III's fsm pass marked any word; control is 1 when any label row has role control or mixed; and wide buses counts the distinct bus names with a bit index of at least 1. The matrix uses one symbol per cell: ✗ for count 0, ✓ for one, ✓✓ for two or more; FSM and control are 0/1 flags, so they show ✗ or ✓ only.

The bus column is buses_wide, not buses. The latter counts every named signal, including single-bit ports, so it is at least 1 for all 150 designs and at least 2 for 147, and would put a check mark in every row. The wide count is the quantity a "multi-bit bus" row means; both columns are committed in features.csv.

Table IV shows 12 designs, three per quartile at fixed ranks by size, so the sample is not hand-picked; Table V counts, per quartile, the designs that show each feature. The storage trend is partly tautological: a clocked block adds 1000 to the score (Eq. 3) and storage counts register words, so storage rises with the quartile by construction; the other trends are conditioned on the score as well. Storage is absent from Q1 and Q2 (0 of 77 designs) and present in 31 of 37 Q3 and all 36 Q4 designs. The other features grow with the quartile: select/mux 8% in Q1 to 56% in Q4, control 3% to 83%, FSMs 0 of 77 below Q3 and 15 of 36 in Q4. Arithmetic is rare throughout (21 of 150, 14%), so the benchmark leans on control, selection and storage more than on arithmetic datapaths. Wide buses are common from Q2 up (90%, 59%, 100%) but nearly absent in Q1 (8%), whose members mostly have only single-bit signals. The FSM flag is Odin III's detection, not a reading of the source: Prob107_fsm1s in Table IV has storage and control cells but no FSM mark.

D. Benchmark size per technology mapping

Table VI gives the size of the 150 designs on each floor. The averages hide a heavy tail. Over all designs the mean node count is 163.5 on f1-k6, 111.4 on f2 and 362.0 on f3, but the medians are 14.0, 13.5 and 25.0: most references are small (125 of 150 have at most 100 nodes at f3) and a few are large (9 exceed 1000; the largest, Prob144_conwaylife, has 15,487). The gate

TABLE III
The VerilogEval spec-to-rtl references by Gate-Breaker complexity quartile (absolute bins of Table II; score range observed in the 156 refs; pyverilog AST scores against regex-fallback scores; built = designs with all three floors).

Quartile	Bin	Observed	Refs	AST/regex	Built
Q1	3–23	3–23	38	36/2	38
Q2	24–111	24–90	39	23/16	39
Q3	112–1064	120–1060	40	1/39	37
Q4	1065–1326	1069–3259	39	0/39	36
All			156	60/96	150

TABLE IV
Benchmark feature matrix, a sample of 12 of the 150 designs (the designs at ranks $\lfloor n/6 \rfloor$, $\lfloor n/2 \rfloor$ and $\lfloor 5n/6 \rfloor$ by f3 node count inside each quartile). ✗: feature absent; ✓: one; ✓✓: more than one; FSM and control are 0/1 flags. Features are read from the f3 floor (Section IV-C).

Design	Q	f3 nodes	Arith	Sel	Stor	FSM	Ctl	Bus
Prob028_m2014_q4a	Q1	3	✗	✗	✗	✗	✗	✗
Prob036_ringer	Q1	8	✗	✗	✗	✗	✗	✗
Prob091_2012_q2b	Q1	17	✗	✗	✗	✗	✗	✓
Prob125_kmap3	Q2	15	✗	✗	✗	✗	✗	✗
Prob015_vector1	Q2	32	✗	✗	✗	✗	✗	✓✓
Prob123_bugs_addsubz	Q2	166	✓✓	✓	✗	✗	✗	✓✓
Prob107_fsm1s	Q3	11	✗	✓	✓	✗	✓	✗
Prob117_circuit9	Q3	28	✓	✓	✓✓	✗	✓	✓
Prob045_edgedetect2	Q3	89	✗	✗	✓✓	✗	✗	✓✓
Prob128_fsm_ps2	Q4	30	✗	✗	✓	✓	✓	✓✓
Prob140_fsm_hdlc	Q4	52	✗	✗	✓	✓	✓	✓
Prob115_shift18	Q4	1121	✗	✓✓	✓✓	✗	✓	✓✓

mapping multiplies the node count of the LUT mapping by about 3 on the mean and 1.9 at the median (362.0 against 111.4 on average, 25.0 against 13.5 at the median), as expected when a LUT of up to six inputs becomes a network of cells from a library of one- and two-input gates (BUF, INV, NAND2, NOR2). f1-k6 sits above f2 on average and in Q3 and Q4 (109.4 against 77.4; 453.9 against 269.8) while the two coincide in Q1 and Q2 (6.3 and 6.1; 100.2 and 100.0): the f1-k6 floor keeps hard cells (21 designs hold hard adders) where f2 spreads arithmetic over LUTs, which is consistent with the excess but is not isolated here. By quartile on f3, the mean grows from 9.9 nodes in Q1 to 972.1 in Q4, but the Q4 median is only 51.5: the complexity score tracks the number and kind of HDL constructs, not the netlist size. For scale, the micro suite's f3 floors hold 50,047 nodes over 65 designs, a mean of about 770.

V. Results

A. L0 on VerilogEval: the provenance ceiling

The provenance experiment is run on the 156 VerilogEval spec-to-rtl reference designs (Section IV) next to the micro suite. The question is the one L0 exists to answer: how much of the reference does the answer key reconstruct, per floor, with no recovery step at all. The experiment builds the three floors f1-k6, f2 and f3 for every reference (f1-earch is not built; one architecture is enough because f2 and f3 do not depend on it).

Coverage of the build. Of the 156 references, 150 build on all three floors, and loki check reports zero errors on the assembled dataset. Two are skipped by Odin III's SystemVerilog reader: Prob151_review2015_fsm and Prob156_review2015_fancytimer (both Q4), with ODIN3_ERR_PARSE ... an explicit 4-bit cast is not supported. Four fail Loki's own equivalence gate and are recorded as failed rather than forced in: Prob031_dff (the reference initialises q to 1'hx; the writer-versus-Odin III BLIF check reports differs on f1 and f2, while f3 was equivalent) and Prob046_dff8p, Prob078_dualedge and Prob145_circuit8, whose negedge flip-flops make equiv-check return an error status for both comparisons. One design, Prob144_conwaylife, was killed by Wyrd's default 2 GB step memory limit (peak 4.3 GB) and built after a rerun with an 8 GB limit. Reading .sv sources needed two fixes in Loki (SystemVerilog is read with –nolatches only, since –vtr-compat is refused for it, and the slang adapter is registered with the process); both are in loki/odin.py and apply to the experiment run.

The ceiling. On f3 the 150 designs have 54,306 gate nodes. Every node has a label row, a cell_type and an HDL span: the label, cell_type and HDL coverage have geometric mean 1.000 over designs. Label coverage is rows over nodes, so it shows that a row exists, not that its fields are informative; unmatched cells stay unknown. The 150 module scopes of the 150 reference modules are all present (module coverage 1.000). Signal coverage, the recovered leaf signal names over the signals the reference declares, has geometric mean 0.986 (arithmetic mean 1.029): 120 designs are at exactly 1, 28 are below it, and 2 are above it, where flattened array elements add names. The micro suite shows the same shape with a slightly lower ceiling (Table VII): 99.1% of module scopes and 91.7% of signal names (an aggregate ratio, where VerilogEval's 841 over 745 is inflated by those two designs; capped at each design's declared count it is 715 over 745, 96.0%). The two sets agree on the message of Section V-C: every node has a label row, a cell_type and an HDL span, and what the mapping erased is the signal names, not the label rows. (Label coverage counts rows, not informative values: a cell that no rule matched keeps the value unknown.)

Caveats, reported with the numbers. (i) –nolatches turns inferred-latch paths into don't-care and lower –async sync turns asynchronous resets into synchronous clears, so a built netlist can differ from the reference's behaviour even though Loki's own checks (netlist against its anonymised twin and against Odin III's writer) pass; the build gate guarantees internal consistency, not equality with the reference. (ii) The reference module, instance and signal counts come from pyverilog where it parses the reference (130 of the 150) and from a regex fallback otherwise (20); the regex counts are approximate and the reference signal count is the weakest number in

TABLE V
Designs showing each feature, per complexity quartile: count (percent of the quartile). The last column counts designs with two or more instances of the feature (always 0 for the 0/1 flags). Wide buses are distinct multi-bit buses; a design with only single-bit ports has none.

Feature	Q1 (n=38)	Q2 (n=39)	Q3 (n=37)	Q4 (n=36)	All (n=150)	≥ 2
Arithmetic	2 (5%)	4 (10%)	7 (19%)	8 (22%)	21 (14%)	9
Select/mux	3 (8%)	10 (26%)	12 (32%)	20 (56%)	45 (30%)	12
Storage	0 (0%)	0 (0%)	31 (84%)	36 (100%)	67 (45%)	12
FSM	0 (0%)	0 (0%)	1 (3%)	15 (42%)	16 (11%)	0
Control	1 (3%)	4 (10%)	15 (41%)	30 (83%)	50 (33%)	0
Wide buses	3 (8%)	35 (90%)	22 (59%)	36 (100%)	96 (64%)	55

TABLE VI
Benchmark size per technology-mapping floor, 150 designs: mean / median node count (top), and per complexity quartile (bottom; nodes). Nodes are all graph nodes of the floor netlist: primary inputs and outputs, constants, cells.

All 150 designs	f1-k6	f2	f3
Nodes	163.5 / 14.0	111.4 / 13.5	362.0 / 25.0
Cells	114.6 / 5.0	62.5 / 4.0	313.2 / 13.5
Nets	218.3 / 12.0	93.5 / 10.0	344.2 / 21.0
Nodes by quartile	**f1-k6**	**f2**	**f3**
Q1 (n=38)	6.3 / 5.0	6.1 / 5.0	9.9 / 8.0
Q2 (n=39)	100.2 / 26.0	100.0 / 26.0	238.2 / 32.0
Q3 (n=37)	109.4 / 18.0	77.4 / 13.0	260.6 / 28.0
Q4 (n=36)	453.9 / 26.5	269.8 / 26.5	972.1 / 51.5

TABLE VII
The L0 ceiling on f3, micro suite and VerilogEval references. Both columns are counts over all built designs; the last row is the per-design coverage of declared signals.

	Micro	VerilogEval
Source designs	68	156
Built on f3	65	150
Refused by the reader	3	2
Failed equivalence gate	0	4
Gate nodes, each with a label row	50,047	54,306
Rows with cell_type and HDL span	100%	100%
Module scopes / reference modules	112 / 113	150 / 150
Leaf signal names / declared signals	611 / 666	841 / 745
Per-design signal coverage: median	100%	100%
minimum	55.6%	75.0%
geometric mean	–	0.986

this section. (iii) The same pyverilog limit decides the complexity quartiles (Section IV-B). (iv) A train/test split by design is fixed by the manifest rule (117 train, 33 test on f3); it is not used here, where no model is fitted.

B. L1 recovery: what the flow already reverse engineered, and what it erased

Table VIII reports L1 on the renamed floors (identical to base, as the neutrality check requires). The head-line result is the hard detector: hard adders recover 67 of 67 with precision 0.985, hard multipliers 30 of 30, because the synthesis flow itself already reverse engineered them and unmap says so. The rest is the honest difficulty: on f1 the calibrated macro-F1 is 0.354 (counted 271, matched 107, ground truth 688), with ff_en at 0.364, registers at 0.021 and muxes at 0.018; on f2 it is 0.108 (91/19/686), with adders at F1 0.237 (precision 1.000, recall 0.134) and most other kinds near zero. Without calibration the flow proposes far more and confirms little: f1 macro-F1 0.202 with 1,128 counted (323 of them declined-adder proposals), f2 0.075. Calibration is honest about itself: ECE 0.217 calibrated against 0.650 uncalibrated.

TABLE VIII
L1 recovery on the renamed floors, per kind. Precision/recall/F1 against the L0 ground truth; calibrated (micro-fit) columns. The hard kinds recover from unmap facts; the rest must be re-found from the gates.

Kind (f1)	Precision	Recall	F1
hard_add	0.985	1.000	0.993
hard_mul	1.000	1.000	1.000
ff_en	–	–	0.364
register	–	–	0.021
mux	–	–	0.018
mux_bit	–	–	0.081
macro-F1 (all kinds)			0.354
Kind (f2)			
adder	1.000	0.134	0.237
subtractor	1.000	0.074	–
mux_bit	–	–	0.100
adder_cin	–	–	0.029
macro-F1 (all kinds)			0.108

C. f3: recovery from the cmos6 gate netlist

Gate-Breaker's own input is a cmos6 gate netlist, so we built the f3 floor (65 of 68 micro designs; the gate reader refuses three hard-block sources) and asked the same questions of it. The L0 answer key is complete at this level in that all 50,047 gate nodes carry a label row, a cmos6 cell_type and an HDL span. What provenance reconstructs of the reference is nearly complete but not total: 112 module scopes against 113 module definitions (99.1%; 54 of 65 designs complete) and 611 distinct leaf signal names against 666 declared signals (91.7%; median design 100%, minimum 55.6%). The module gap is reference-side (generate branches and unused submodules never elaborate); the signal gap is synthesis-side (nets that mapping created or merged have no original name left). The ceiling is consequently what the mapping preserved, not the source RTL.

TABLE IX
f3 L1 per kind on the 15 test designs. The calibrated columns are the protocol's setting; the –calib none columns are a diagnostic of what the detectors propose before calibration prunes it. A kind with no counted prediction has no F1 and is left out of the macro average.

		calibrated			–calib none			
Kind	GT	counted	matched	F1	counted	matched	precision	F1
adder	17	0	0	–	1	1	1.000	0.111
adder_cin	17	0	0	–	0	0	–	–
comparator	20	0	0	–	14	0	0.000	0.000
eq_const	16	0	0	–	1,141	0	0.000	0.000
ff_srst	2	0	0	–	7	0	0.000	0.000
hard_add	17	0	0	–	0	0	–	–
hard_mul	11	0	0	–	0	0	–	–
mux	7	0	0	–	0	0	–	–
mux_bit	7	0	0	–	0	0	–	–
register	22	0	0	–	3	1	0.333	0.080
subtractor	8	0	0	–	0	0	–	–
macro-F1	144	0	0	0.000	1,166	2		0.038
buses	120	2,569	0	0.000	2,835	0	0.000	0.000
datapath role	20,541	20,541	20,541	1.000	20,541	20,541	1.000	1.000
control role	0	0	0	–	0	0	–	–

TABLE X
f3 AST (Gate-Breaker Algorithm 2, depth 3, pyverilog) and CFG (Algorithm 3, odin3 probe) similarity against the micro reference RTL, L0-stop (gate netlist) against L1-stop (design.v); quartiles by f3 node count. With the f3 calibration L1-stop equals L0-stop, so every delta is 0.0000; the table therefore shows the L0 values and the diagnostic (–calib none) deltas.

Quartile	AST L0	AST L1	ΔAST	CFG L0	CFG L1	ΔCFG
0 (4 designs)	0.4428	0.4428	0.0000	0.7909	0.7909	0.0000
1 (4)	0.3765	0.3765	0.0000	0.4224	0.4224	0.0000
2 (4)	0.3969	0.3969	0.0000	0.7013	0.7013	0.0000
3 (3 AST / 1 CFG)	0.4003	0.4003	0.0000	0.7507	0.7507	0.0000
mean (15 / 13)	0.4044	0.4044	0.0000	0.6468	0.6468	0.0000
diagnostic –calib none (L1 commits hierarchy)						
0 (4)	0.4428	0.4277	−0.0151	0.7909	0.7776	−0.0133
1 (4)	0.3765	0.3756	−0.0009	0.4224	0.4099	−0.0124
2 (4)	0.3969	0.4542	+0.0573	0.7013	0.7014	+0.0001
3 (3 AST / 1 CFG)	0.4003	0.4007	+0.0004	0.7507	0.7561	+0.0055
mean	0.4044	0.4155	+0.0111	0.6468	0.6394	−0.0075

Table IX reports L1 on the 15 f3 test designs (23,098 gate nodes), base and renamed, with a calibration fitted on the 50 train designs' f3 floors. The result is the strict one: no construct is counted (0 of 144 ground-truth instances; macro-F1 0.0) and no hypothesis is committed at all. The f3 priors are strongly negative because the gate-level detectors are imprecise on the train floors (comparator −9.24 from 31,036 proposals with 2 true), so the best calibrated confidence in any test design is a bus at $p = 0.764$, below the 0.9 commit threshold, and the L1-stop Verilog is byte-identical to the input netlist in 15 of 15 designs. Without calibration the detectors count 1,166 constructs for 2 matches (macro-F1 0.038; one adder, one register) and 2,835 buses for 0 matches. Buses are proposed but never reconstructed: 2,569 counted bus hypotheses against 120 ground-truth buses, 0 matched, so no width or bit order is recovered. The role detector labels every logic cell datapath, which is 100% correct only because the f3 answer key holds no control cells (0 of 50,047; f1 has 4): control-versus-data is not answerable from f3 provenance, and the datapath F1 of 1.000 over 20,541 cells is vacuous. The hard detector stays f1-only by design (unmap runs only on f1 floors with hard cells), so the 28 hard ground-truth instances are unreachable here. Base and renamed score.json digests are identical (9368cfc26303): naming contributes no decision bit at f3 either.

The AST and CFG deltas (Table X) follow from the zero-commit result: with the f3 calibration L1-stop is byte-identical to L0-stop, so both metrics have delta exactly 0.0000 in every quartile (AST over 15 designs; CFG over the 13 whose CFG the probe can finish, 12,000 nodes being its practical $O(n^2)$ limit). The uncalibrated diagnostic, where L1 commits 1,535 hypotheses in 11 of 15 designs, moves the means by +0.0111 (AST) and −0.0075 (CFG), with mixed per-quartile signs; the commits are mostly wrong eq_const and bus pieces, so this is not evidence about recovery quality. A flatten control (Odin III proc; flatten, which also lowers the committed processes) inlines the committed hierarchy and raises both metrics over the gate netlist: AST 0.4576 against 0.4044 (+0.0533, all 15 designs) and CFG 0.6941 against 0.6382 (+0.0559, the 12 designs the probe can score). Gate-Breaker's Algorithms 2 and 3 read a single module, while Loki's L1 commits a hierarchy, so a recovery that produces hierarchy is systematically penalised until it is flattened: for a netlist-to-RTL benchmark this is a metric limitation, not a property of the recovery.

D. L2: clusters that carry the hierarchy, and a GNN that beats 1-WL

Table XI scores the four clusterings against the known module hierarchy. Louvain leads (modularity 0.4220, NMI 0.1360); k-means and affinity propagation do not find the hierarchy (negative modularity), which is itself a result: graph structure alone, without the labels, favours the community methods. 271 of 272 partition records are identical between base and renamed floors; the one difference exposed a determinism defect in the spectral path (repeated runs on the same base design give modularities -0.0079, -0.0410 and 0.0252), reported as an open issue rather than hidden.

The GNN result answers the paper's question directly. A node-only GraphSAGE loses to the 1-WL baseline on real micros (macro-F1 0.2413 against 0.2711); combining node, graphlet, WL and NPN features turns it around (0.3545 against 0.2711 on the earlier real-micro run), and the full labelled run of this experiment gives test macro-F1 0.4241 against 0.2717 (accuracy 0.8863 against 0.8492, 3,932 test nodes across 13 designs). The features carry the recovery, not the message passing alone.

E. The Gate-Breaker metrics

Gate-Breaker's metrics compare a recovered RTL against a reference: Success Rate, Success@k, AST similarity and CFG similarity. Loki recovers structure, not a new RTL text, so the direct comparison is the replication: the Gate-Breaker harness itself, run on the simple benchmarks with the only language model available on the host, gave

TABLE XI
L2 clustering against the known module hierarchy (68 designs, mean over the renamed floors, identical to base). Modularity and NMI(module_path) per method.

Method	Modularity	NMI
Louvain	0.4220	0.1360
Spectral	0.3620	0.1378
Affinity propagation	-0.2264	0.2708
k-means	-0.2912	0.3220

TABLE XII
Gate-Breaker metrics on the VerilogEval quartiles (one run, K=10, 4 workers, 180 s per call, 7,440 s wall clock, 740 calls; deltas vs the paper's O3-Mini in the text).

Quartile (n)	SR	S@1	S@5	S@10	AST	CFG
Q1 (36)	0.903	0.944	1.000	1.000	0.894	0.942
Q2 (37)	0.778	0.811	1.000	1.000	0.761	0.890

success rate 0.933 on Q1 and 0.829 on Q2, Success@10 1.000 in both quartiles, AST similarity 0.576 and 0.444, CFG similarity 1.000 and 0.736, with 169 of 174 extracted candidates functionally correct – a first replication on the fallback micro set, not the paper's VerilogEval set. The VerilogEval replication, on the quartiles the paper used, is reported in Table XII with the AST and CFG kernels recomputed with common-width feature vectors (the first run padded per graph and truncated in the zip, which the v1.1.0 CFG probe found; those columns are not comparable to it).

Seventy-four of the 78 Q1+Q2 designs are usable (three excluded because their synthesized netlists fail the official testbench self-check, one because its 241 KB prompt exceeds the OS argument limit); every scored design is solved by attempt 5. The deltas against the paper's O3-Mini: Q1 success rate -0.092, AST -0.030, CFG -0.011; Q2 success rate -0.039, Success@5 +0.125, Success@10 +0.111, AST -0.164, CFG +0.083. The Q1 gap is harness no-output, not wrong RTL; the Q2 AST gap is truth-table and case references against the model's boolean expressions. Two caveats: the host's only model path routes through a gateway that aliases a DeepSeek backend as Claude Opus (so the model column is not verifiably the paper's comparison class), and the CLI exposes none of the paper's sampling flags (T=0.85, p=0.95). The CFG column uses the common-width node kernel (the odin3 CFG probe); the AST column uses the paper's pyverilog Algorithm 2 of Gate-Breaker.

What the two lines of work measure is complementary. The language model recovers functional behaviour from the gates and names nothing reliably; Loki recovers the structures the flow preserved (hard blocks, the hierarchy, the signal roles its features see) and proves what it claims. The Gate-Breaker future work – bus naming, datapath/control classification, partitioning – is exactly L2's table, and its numbers are the baseline a model-assisted Loki would have to beat.

TABLE XIII
Round-2 L1 (RQ-L1): recovery from the f3 gate netlist re-mapped to LUT6 against the native f2 floor, same 65 micro designs. test is the 15 held-out designs, all the 65. Calibration: the micro-fit f2 table (gate rows: the f3 table). Hard kinds excluded from counts and averages; macro-F1 averages the kinds with a counted prediction.

Netlist	Calib.	Split	Runs	Counted	Matched	GT	micro-F1	macro-F1
f3 gate netlist	calibrated	test	15	0	0	116	0.000	0.000
	uncalibrated	test	15	1,166	2	116	0.003	0.038
f3 re-mapped to LUT6	calibrated	test	15	68	2	116	0.022	0.076
	calibrated	all	65	92	14	580	0.042	0.062
	uncalibrated	test	15	252	4	116	0.022	0.044
	uncalibrated	all	65	921	31	580	0.041	0.049
f2 floor (native LUT6)	calibrated	test	15	72	4	116	0.043	0.153
	calibrated	all	65	88	16	579	0.048	0.095
	uncalibrated	test	15	255	13	116	0.070	0.121
	uncalibrated	all	65	930	32	579	0.042	0.070

TABLE XIV
Operator-recovery F1 (RQ-L7) of the existing and new L1 runs: the multiset of recovered operators (class and operand width) against the operators of the L0 key, pooled over designs. type ignores the width, tw requires it.

Run set	Designs	Key ops	Got ops	P	R	type-F1	tw-F1
f1-k6, calibrated	68	468	169	0.657	0.237	0.349	0.204
f1-k6, none	68	468	964	0.233	0.481	0.314	0.166
f2, calibrated	68	468	21	0.810	0.036	0.070	0.061
f2, none	68	468	756	0.180	0.291	0.222	0.080
f3 gates, calibrated	15	108	0	–	0.000	0.000	0.000
f3 gates, none	15	108	1,161	0.009	0.102	0.017	0.008
f3 as LUT6, calibrated	65	463	19	0.895	0.037	0.071	0.062
f3 as LUT6, none	65	463	753	0.190	0.309	0.235	0.086
VerilogEval f2, calibrated	150	3,585	11	0.727	0.002	0.004	0.002
VerilogEval f2, none	150	3,585	1,151	0.196	0.063	0.095	0.042
VerilogEval f3, calibrated	150	3,585	2	1.000	0.001	0.001	0.001
VerilogEval f3, none	150	3,585	5,488	0.107	0.163	0.129	0.027

F. Round 2: five follow-up questions

The first round left five questions open. Each carries a hypothesis and a decision rule; the thresholds were set by the implementer during the round and are not time-stamped before the data (Appendix A); a later round should fix them in a commit before running. All five are compute-only: no language-model call was made, and the language-model figures are the round-1 Gate-Breaker replication, read again. Every number in this subsection is a macro generated from the summary tables of experiments/round2.

RQ-L1: is the f3 failure a vocabulary problem? The L1 detectors read LUT covers, and the f3 netlist is a cmos6 gate netlist. H1: re-mapping the same gates into LUT6 recovers close to the f2 rate. We read each f3 netlist.blif into Odin III, run abc –lut 6, and give the result to loki re as an f2 netlist. Odin III drops the cell names in abc, so each of the 8,712 LUTs is expanded back to the gates it covers: a bit-parallel simulation signature matches its output to the f3 net of the same function, and the cover is the cone of that net down to the matched inputs. 0.230 of the LUTs ABC resynthesised implement a function that is no f3 net; they take the support-contained share of their consumers' covers, and 153 are left with an empty cover. The expanded hypotheses are then scored against the unchanged f3 key, and the native f2 floor of

TABLE XV

AST and CFG similarity against operator F1 on the 15 f3 test designs (the f3-recovery runs). Calibrated and uncalibrated L1 tie on AST and CFG; the flatten control moves both and leaves the recovery record unchanged.

Condition	AST	CFG	Ops recovered	type-F1
L0-stop gate netlist	0.4044	0.6468	0	0.0000
L1 calibrated	0.4044	0.6468	0	0.0000
L1 uncalibrated	0.4155	0.6394	1,161	0.0173
L1 uncalibrated, flattened	0.4576	0.6941	1,161	0.0173

TABLE XVI

Head-to-head on the VerilogEval designs the language model was scored on (Q1 and Q2, functionally correct samples; AST depth 3 and CFG as in Table XII). Loki columns: the output Verilog of the flow at that floor against the reference .sv; L0 is the netlist before L1.

Metric	LLM (round 1)	f3 netlist (L0)	f3 L1 cal.	f3 L1 none	f2 L1 cal.	f2 L1 none
Q1 AST	0.894	0.763	0.763	0.680	0.804	0.720
Q1 CFG	0.942	0.751	0.751	0.565	0.821	0.664
Q2 AST	0.761	0.460	0.460	0.438	0.459	0.448
Q2 CFG	0.890	0.743	0.743	0.679	0.787	0.764
Q1+Q2 AST	0.827	0.611	0.611	0.559	0.632	0.584
Q1+Q2 CFG	0.916	0.747	0.747	0.622	0.804	0.714

TABLE XVII

Loki on all 150 VerilogEval designs by complexity quartile (mean AST and CFG of the output Verilog against the reference; last column: operator type-F1 of the uncalibrated f2 run). Calibrated f3 equals the f3 netlist wherever L1 commits nothing.

Quartile	n	f3 AST	f3 CFG	f3 cal. AST	f3 cal. CFG	f2 cal. AST	f2 cal. CFG	op F1
Q1	38	0.754	0.750	0.754	0.750	0.793	0.820	0.061
Q2	38	0.455	0.741	0.455	0.741	0.455	0.784	0.075
Q3	32	0.614	0.751	0.605	0.750	0.607	0.811	–
Q4	21	0.505	0.768	0.505	0.768	0.507	0.870	–

TABLE XVIII

Evidence-family ablation (RQ-L4) on the 49 held-out f2 designs (16 micro, 33 VerilogEval). Each row refits the calibration on the 169 train designs with the family removed (its weight set to 0). The l2 family has no feature in loki re and is not applicable. The interval is a 95% design bootstrap of micro-F1 (hard kinds excluded); the next column is micro-F1 with the design ve_Prob153_gshare (about 1,300 of the 1,558 instances) left out.

Calibration	Counted	Matched	micro-F1	95% interval	w/o gshare	macro-F1	ECE	AUC
defaults (no calibration)	1,708	26	0.0159	0.0036–0.0912	0.0553	0.113	0.841	0.437
full calibration	10	3	0.0038	0.0000–0.0444	0.0225	0.145	0.061	0.885
without npn	10	3	0.0038	0.0000–0.0444	0.0225	0.145	0.061	0.885
without wl	13	3	0.0038	0.0000–0.0435	0.0222	0.145	0.024	0.930
without struct	11	4	0.0051	0.0000–0.0578	0.0299	0.163	0.056	0.884
priors only	14	4	0.0051	0.0000–0.0574	0.0295	0.163	0.026	0.918

the same 65 designs, with the same calibration, is the comparison (Table XIII). On the gate netlist the calibrated flow matches 0 of 116 instances in the 15 test designs. After the LUT6 re-map it matches 2 (native f2: 4), and over all 65 designs 14 against 16 of 580, a micro-F1 ratio of 0.87 in-sample (the 65 designs include the designs the f2 calibration was fitted on) against 0.51 on the held-out test split. The rule asked for a ratio of at least 0.8 on both splits, so H1 is inconclusive. The in-sample recovery is that of f2, on the same kinds (adders, subtractors, muxes), but the 15-design test split holds only a handful of matches (2 against 4), so its ratio is noise in either direction. The result is consistent with the f3 failure being the detectors' vocabulary and not missing information in the gates, but does not show it; the f2 level it would recover to is itself low.

RQ-L7: are AST and CFG blind to what Loki recovers? We score recovery with a word-level operator F1. The L0 key lists the word cells of the elaborated reference; a run's recovered operators are its counted hypotheses (committed, or $p_{\text{comp}} \geq 0.5$), mapped to an operator class and an operand width (add, sub, mul, eq, ne, lt, le, gt, ge, mux, register). Precision and recall are multiset overlaps, so the score does not depend on which cells carry an operator; type ignores the width and tw requires it. Table XIV rescores every run set. The floors separate cleanly, which neither AST nor CFG can show because both exist only on f3: type-F1 0.349 on f1 (the hard detector), 0.070 on f2 and 0.000 on the f3 gates. On the f3 test designs (Table XV) the calibrated and uncalibrated L1 runs tie on both metrics, AST by 0.0111 and CFG by -0.0075, while operator type-F1 moves by 0.0173; the flatten control raises AST by 0.0422 and CFG by 0.0547 and leaves the recovery record, hence the operator F1, unchanged (0.0000). By the rule H7 is supported by the letter of the rule (weak evidence). The tie the rule was written for, the calibrated flow against the L0-stop netlist (both add nothing to the text), does not separate: their operator type-F1 is 0 against 0. The verdict rests on the calibrated flow against the uncalibrated one, where the calibrated side recovers no operator at all, and on the flatten clause, which holds by construction. Two cautions: the separating level is small (the uncalibrated detectors count mostly wrong constructs), and the flatten comparison holds the record fixed by construction, because the operator F1 reads the recovery record and AST and CFG read the text. The operator F1 does not replace them; it is the measure that sees recovery, where they see only form.

RQ-L2: Loki against the language model on the same designs. Round 1 scored the language model on Q1 and Q2 of VerilogEval, and Loki's AST and CFG only on micro designs. We ran loki re on all 150 VerilogEval netlists at f2 and f3, calibrated (a calibration fitted on the 117 train designs of VerilogEval itself) and not, and scored the output Verilog (design.v) against the reference .sv with the round-1 AST and CFG code. 21 references that pyverilog cannot parse have no AST value for any candidate, and 6 f3 outputs exceed the CFG probe's size limit (l2_coverage.csv). On the designs both are scored on (Table XVI), Loki's f3 output has AST similarity 0.763 in Q1 and 0.460 in Q2 (common designs, $n = 36$ and 36) against the model's 0.894 and 0.761, and CFG similarity 0.751 and 0.743 against 0.942 and 0.890; every gap is significant by a paired Wilcoxon test (l2_paired.csv). The calibration was fitted on the train designs and these means cover all 150, so we also give the held-out test split alone (l2_testsplit.csv): on its 16 Q1 and Q2 designs both tools

TABLE XIX
Graph sandwich (RQ-L5): Spearman correlation of the bracket width (candidate fraction of the cells) and of the committed fraction with the per-design F1; cell-level soundness against the union of the scored ground-truth footprints (must: committed cells; may: committed plus candidate cells).

Run set	Designs	ρ width	p	ρ must	must prec.	must rec.	may rec.
VerilogEval f2, uncalibrated	90	0.087	0.417	0.485	0.840	0.240	0.625
VerilogEval f2, calibrated	90	−0.203	0.055	0.819	0.821	0.025	0.188
VerilogEval f2, calibrated, rp 0.1	90	−0.346	0.001	0.819	0.821	0.025	0.447
VerilogEval f3, uncalibrated	90	−0.124	0.243	0.168	0.792	0.347	0.727
micro f2, uncalibrated	48	0.162	0.272	0.219	0.879	0.268	0.364
micro f2, calibrated	48	0.052	0.725	0.844	0.932	0.021	0.132

TABLE XX
Structural prefilter (RQ-L5b), measured offline on the 210 cone_check calls of all round-2 runs (166 proved equivalent): the rules that prune calls without losing a proved construct, and the largest rule overall.

Rule (kind/feature=bin)	Calls pruned	Fraction	Equivalent lost
adder/out_reads=no	14	0.067	0
subtractor/width=2	8	0.038	0
subtractor/out_reads=no	4	0.019	0
adder/npn_sum=low	3	0.014	0
subtractor/npn_carry=all	3	0.014	0
unrestricted best	79	0.376	79

score on, AST is 0.705 for Loki against 0.893 for the model and CFG 0.802 against 0.927, the same direction. (The calibrated f3 flow commits in only 7 designs, so the fit can change few outputs.) Across the four quartiles (Table XVII) the f3 AST over all scored designs ($n = 38$, 38, 32, 21; a different, larger design set than the common rows above) reads 0.754, 0.455, 0.605 and 0.505, so it falls from Q1 to Q4 without being monotone. The 21 references that do not parse fall in Q1 to Q4 as 0, 1, 5 and 15, so Q3 and Q4 are scored on the designs that parse, and the apparent fall may partly reflect which designs those are. The CFG stays flat (0.750 to 0.768). The calibrated f3 flow commits almost nothing, so its output is the L0 netlist; the uncalibrated flow, which does commit, scores lower, not higher.

What the correctness column proves is narrow. Every one of the 150 runs of each set ends final_equiv = equivalent (150 for the calibrated f3 set), which says that the flattened output is formally equal to the input netlist. It does not say the output matches the reference RTL: that netlist itself passes the official testbench in only 128 of the 150 designs (the round-1 self-check; inferred latches and asynchronous resets are the failures), so Loki's output is exactly as correct as its input, no more. The model is checked against the testbench directly, but only its correct samples enter the similarity columns. On operator recovery (RQ-L7) the model is far ahead: on the common Q1 and Q2 designs its operator type-F1 is 0.465 (functionally correct samples elaborated with the same Odin III front as the key) against 0.081 for Loki's calibrated f2 run and 0.074 uncalibrated (the calibrated f3 run recovers no operator, 0.0, so the comparison uses f2, a different floor from the one AST and CFG use). By the rule, H2 is inconclusive: Loki is below the model on both metrics in both quartiles (yes), the equivalence holds in all runs (yes), and the similarity does not stay flat from Q1 to Q4 (no). The positioning this measures is that the model is readable and recovers the operators of small designs but is unverified, and Loki is verified but partial; neither number alone is the result, and the arm that combines them (a model given the ReverseProvenance record, L8) was not run.

RQ-L4: which evidence family drives the posterior? The calibrated flow scores every hypothesis from a per-detector prior plus feature log-likelihood ratios in four families (npn, wl, l2, struct). loki re produces no l2 feature (those exist only in the ReverseProvenance document), so that family is not applicable. For each remaining family we refit the calibration on the train designs of the micro and VerilogEval f2 floors with the family removed and its weight set to 0, and ran the 49 held-out designs (Table XVIII; 1,558 ground-truth instances). Held-out micro-F1 is 0.0159 with the defaults, 0.0038 with the full calibration and 0.0051 with priors only. The full-minus-priors gap is −0.0013, below the 0.02 floor of the rule, so H4 is inconclusive: no family carries measurable performance, because the features add nothing detectable over the per-detector priors. One held-out design (ve_Prob153_gshare) holds most of the ground truth and nothing in it is matched; with it left out (257 instances) the micro-F1 is 0.0553 with the defaults, 0.0225 full and 0.0295 priors only (gap −0.0070), so the verdict does not change; on that subset the defaults score higher than either calibrated row. The fit says so itself: the full calibration sets the npn weight to 0 and shrinks struct and wl (l4_weights.csv). What calibration does is prune, from the many counted constructs of the defaults to a handful, and rank: the AUC of p_{comp} over the live hypotheses rises from 0.437 to 0.885 (over different pools, since calibration rejects more).

RQ-L5: is the must/may bracket an uncertainty measure? Following the graph-sandwich framing (must is what L1 proved and committed, may adds the candidates it lists), the bracket of a run is read from reverse-provenance.json and its width is the candidate fraction of the cells. H5a: the width correlates negatively with the design's F1. On the uncalibrated VerilogEval f2 runs (90 designs with ground truth) the Spearman correlation is 0.087 ($p = 0.417$), and −0.346 on the calibrated run that lists candidates from $p = 0.1$ (Table XIX), so H5a is not supported. The committed fraction does correlate with F1 (0.485), but trivially: committed cells are the matches. As a soundness check of the bracket itself, 0.840 of the committed cells lie inside the scored ground-truth footprints (must precision), the committed cells cover 0.240 of the truth, and the may bracket contains 0.625 of it. H5b: of 210 cone_check (SAT) calls across all round-2 runs, 166 end equivalent; the largest structural rule that loses no such verdict (adder/out_reads=no) prunes 0.067 of the calls, against the 0.3 the rule asked for (Table XX),

TABLE XXI
RQ-L9, primary set P (M2N+V2N, f2, uncalibrated): counted candidates, false ones, and the share of the false ones per class (rules of EXPERIMENT.md).

kind	cand.	false	same ext.	frag. rel.	frag. other	straddle	partial	noise
ff_en	1,070	1,067	0.01	0.99	0.00	0.00	0.00	0.00
mux_bit	344	336	0.00	0.61	0.39	0.00	0.00	0.00
eq_const	1,105	1,096	0.00	0.18	0.82	0.00	0.00	0.00
register	155	116	0.03	0.64	0.15	0.19	0.00	0.00

TABLE XXII
RQ-L9 (H9b), set P: ground-truth instances with at least two cells, the share whose fragment union covers at least 0.8 of them (q), the same with candidates of any kind, and the mean fragment coverage.

kind	GT	matched	q frag.	q any	mean cov.
ff_en	156	3	0.122	0.160	0.618
mux_bit	368	2	0.005	0.136	0.016
eq_const	995	5	0.040	0.112	0.089
register	289	39	0.194	0.228	0.497

so H5b is not supported. The prover budget is spent confirming true hypotheses, and the structural features the calls already carry do not separate the rest; the truth-table checks (2,876) are cheap already. The rule search covers single-bin rules (one feature and one bin of one kind) on the same data, so 0.067 is an upper bound for such rules, not a measured saving; the calls repeat across the eight run sets. The bound that holds for any rule is simpler: a rule that loses no equivalent verdict can prune at most the 44 calls that were not (0.21 of all).

G. Round 3: fragments or noise, and where the words are

Round 2 left a puzzle. Uncalibrated, the detectors propose many candidates that almost never match (Table XXI); calibrated, they commit almost nothing. Peter's question (Appendix A) is why netlists are hard to bring back to HDL when the structure is there to see: muxes, wide signals that behave as words, state feedback with decode and select, and input and output words that mark where the datapath lies. Round 3 is compute-only on the round-2 runs and the L0 ground truth (no language-model call, no new detector). Unlike round 2, its rules were committed before any number was computed (commit 5f8de24, experiments/round3/EXPERIMENT.md); every figure below is a macro generated from the summary CSVs of experiments/round3. The round-2 counts that motivate the question (for instance ff_en 732 candidates and one correct on the held-out designs) were known when the thresholds were set, so H9a was not blind to the data, only to the round-3 numbers. Analyses added after the data was seen are labelled exploratory, are listed in a dated entry of EXPERIMENT.md, and decide nothing. We added the runs the round-2 sets lacked (micro f3, calibrated and not, and micro f1-k6, uncalibrated; one micro f3 design timed out) and score every set, but the decisions use the primary set P: the micro and VerilogEval f2 runs without calibration.

TABLE XXIII
RQ-L9 for every run set (four decisive kinds pooled): runs, candidates, false candidates and the share that are related fragments, other-type fragments, noise.

set	runs	cand.	false	frag. rel.	frag. other	noise
M2N	65	705	686	0.28	0.69	0.00
M2C	65	71	65	0.95	0.05	0.00
V2N	150	1,969	1,929	0.69	0.30	0.00
V2C	not run					
V3N	150	5,005	4,997	0.29	0.71	0.00
V3C	150	2	0	0.00	0.00	0.00
M3N	64	1,406	1,405	0.23	0.77	0.00
M3C	not run					
M1N	68	496	469	0.38	0.57	0.00

TABLE XXIV
Exploratory, non-deciding additions (2026-10-24, after the data), set P: share of false candidates that are related fragments (by candidate, by distinct cell), the noise′ share (cells only in plain-gate ground-truth words), q over all words and over the words touched by a candidate.

kind	frag. rel.	frag. rel. (cells)	noise′	q	q touched
ff_en	0.99	0.98	0.00	0.122	0.123
mux_bit	0.61	0.61	0.01	0.005	0.006
eq_const	0.18	0.18	0.52	0.040	0.042
register	0.64	0.49	0.00	0.194	0.229
pooled	0.58	0.56	0.22	0.065	0.069

RQ-L9: are the wrong candidates fragments of real words? A false candidate is a counted candidate the one-to-one scorer does not match. We classify each (rules in the protocol): the same extent as a ground-truth instance of its family (a lost tie or a type slip), a fragment of a larger instance of its family, a fragment of an instance of another type, a straddle of several, a partial overlap, or noise. Over the 2,615 false candidates of the four kinds in P, 0.583 (1,525) are fragments of an instance of the matching or a related family. Per kind the related-fragment share is 0.988 for ff_en, 0.638 for register, 0.610 for mux_bit and only 0.175 for eq_const; 3 of four kinds reach 0.5, so H9a is supported by the letter of the rule; not robust. Three qualifications go with it.

First, the noise class is vacuous. The ground-truth instances cover every non-I/O, non-constant cell of the f2 datasets (9,381 of 9,381 in VerilogEval, 9,130 of 9,130 in the micro set), so rule 6 of the protocol could never yield noise, and the noise share 0.000 is structural. The arm of the H9a rule that reads "not supported when the noise share is at least 0.5" was unreachable by construction, and the sentence that no wrong candidate is noise carries no information. As an exploratory replacement we count noise′: a false candidate all of whose cells lie only in plain-gate ground-truth words ($and, $or, $not, $xor, $xnor, the $logic and $reduce gates). Pooled, noise′ is 0.220 (574 candidates); per kind it is 0.522 for eq_const, 0.006

TABLE XXV
RQ-L10, f2 floor, micro and VerilogEval pooled, relaxed rule ($\tau = 0.75$): the share of ground-truth word constructs seeded by a propagated group within h hops of an I/O port word, and the share never reached.

kind	class	GT	$h \leq 1$	$h \leq 2$	$h \leq 3$	$h \leq 4$	never
register	datapath	289	0.57	0.66	0.66	0.69	0.31
mux	datapath	90	0.40	0.41	0.41	0.41	0.59
mux_tree	datapath	13	0.23	0.23	0.23	0.23	0.77
adder	datapath	2,124	0.00	0.00	0.00	0.00	1.00
subtractor	datapath	20	0.20	0.20	0.20	0.20	0.80
comparator	datapath	23	0.30	0.30	0.30	0.30	0.70
counter	datapath	23	0.00	0.00	0.00	0.00	1.00
eq_const	control	990	0.61	0.61	0.61	0.61	0.39
fsm	control	16	0.06	0.06	0.06	0.06	0.94

TABLE XXVI
RQ-L10 by dataset and floor: designs, designs with a port word, input and output port words, and the datapath and control shares reached within two hops.

dataset	floor	designs	with port word	in words	out words	datapath $h \leq 2$	control $h \leq 2$
micro	f2	68	54	158	22	0.35	0.68
micro	f3	65	51	150	29	0.63	0.99
micro	f1-k6	68	54	158	38	0.66	0.78
veval	f2	150	63	103	13	0.07	0.60
veval	f3	150	63	103	13	0.94	0.88

TABLE XXVII
RQ-L11, ground truth: feedback registers, constant-compare decoders on them, sinks (mux selects and enables) and chains, and the muxes with a select (enabled words) whose select (enable) is a chain sink.

dataset	floor	designs	with fb.	with chain	fb. reg.	fsm flag	decoders	sinks	chains	mux S chain/all	enable chain/all
micro	f2	68	3	1	3	0	11	3	12	3/41	0/3
micro	f3	65	3	1	3	0	11	3	12	3/41	0/3
micro	f1-k6	68	3	1	3	0	11	3	12	3/41	0/3
veval	f2	150	55	12	197	16	41	30	89	15/43	15/152
veval	f3	150	55	12	197	16	41	30	89	15/43	15/152

TABLE XXVIII
RQ-L11, detectors: chain elements proposed by a counted candidate with Jaccard at least 0.5 of any kind (in brackets: of the right scored type), chains with all three elements proposed, and ground-truth FSM instances matched by an fsm candidate.

set	fb. register	decoder	sink	complete chains	fsm
M2N	0/3 (0)	1/11 (0)	1/3 (0)	0/12 (0)	0/0
M2C	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0
V2N	41/197 (26)	15/41 (9)	5/30 (0)	5/89 (0)	6/16
V2C	3/197 (0)	1/41 (0)	2/30 (0)	1/89 (0)	0/16
V3N	13/197 (3)	6/41 (2)	3/30 (0)	4/89 (0)	3/16
V3C	4/197 (0)	2/41 (0)	2/30 (0)	2/89 (0)	1/16
M3N	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0
M3C	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0
M1N	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0

for mux_bit, and 0.000 for ff_en and 0.000 for register (Table XXIV). Half of the false eq_const candidates are thus glue logic, not a piece of a word.

Second, the ff_en fragments are near-tautological. An ff_en candidate has two cells (a flip-flop and its enable LUT), and the ground-truth footprint of a register or $dffe word is the whole word, so a correct per-bit slice of a word counts as a fragment of it; 1,054 of the 1,525 related fragments are ff_en. For ff_en the detector finds correct bit slices and the miss is word-level grouping and extent. Without ff_en (exploratory) the pooled related-fragment share over the other three kinds is 0.304 (1,548 false candidates, 2 of three kinds at or above 0.5), and 0.271 when the largest design is also left out. For mux_bit and register the fragment reading holds at the stated shares; for eq_const it is a detector problem: 0.822 of its false candidates are fragments of an instance of another type (mostly single-LUT $and and $or words, registers and fsm instances), so the LUT computes something that looks like a constant compare and is not one. The mux_bit candidates are 0.390 inside an instance of another type ($shiftx words among them).

Third, the result rests on one design. Most ff_en candidates come from ve_Prob153_gshare, where they sit inside a mux tree, and overlapping candidates inflate their count. With that design left out (exploratory, not the rule) 1,863 false candidates remain, 0.415 of them related fragments and 0.563 fragments of another type, so the pooled share falls under the 0.5 of the rule and H9a would be inconclusive; ff_en stays at 0.966 and register drops to 0.333. Counted over the 4,004 distinct (design, cell) pairs covered by false candidates instead of over candidates (exploratory), the pooled related-fragment share is 0.558. We therefore state H9a with both numbers: supported by the letter of the rule at 0.583, not robust (0.415 without ve_Prob153_gshare, 0.304 without ff_en).

H9b asks whether grouping the fragments would recover the word: the share of ground-truth instances whose fragment union covers 0.8 of them is 0.065 pooled (ff_en 0.122, register 0.194, eq_const 0.040, mux_bit 0.005), and no kind reaches 0.5, so H9b is not supported. The denominator holds words no candidate touched; over the 1,698 words touched by at least one candidate (exploratory) the share is 0.069. The shortfall is therefore detector misses plus non-tiling, not one of them: 0.065 of all words, 0.069 of the touched ones. Even with candidates of any kind the cover stays low (register 0.228, mux_bit 0.136). The fragments exist, but they do not tile the words: a merge step over the present detector output would recover few words. The other sets (Table XXIII) have fewer candidates when calibrated, and the gate floor f3 proposes far more false eq_const candidates than any f2 set.

RQ-L10: do the I/O words find the datapath? A top-level port bus is a word the reverse engineer usually has by name. A group of cells is a propagated word if, bit by bit, a known word drives (or is driven by) one cell per bit of the group with the same cell type, truth table and pin, and at least three quarters of the bits qualify (the strict rule asks for all); we propagate for up to four hops. A ground-truth construct is reached when a group of at most h hops lies for the most part inside it. In the f2 floor, 2,582 datapath word constructs (registers, muxes, adders, comparators, counters) and 1,006 control constructs (eq_const and FSMs) have two or more bits; 0.095 of the datapath and 0.605 of the control constructs are reached within two hops, so H10a is not supported

and H10b is not supported. H10b is uninformative by construction once H10a fails: with D = 0.095, "not supported" means $C > D - 0.10 = -0.005$, which holds for any $C \geq 0$. We keep the status the rule gives and do not present it as a finding. Both numbers need reading. The datapath share is pooled and one VerilogEval design alone holds most of the 2,124 adder constructs, which the rule never reaches because the carry chain breaks the bit-for-bit isomorphism; the mean of the per-design shares is 0.326 (datapath, 113 designs) against 0.286 (control, 51 designs; both exploratory), a difference of the sign the hypothesis expects but not of its size. Registers are reached best (0.574 at one hop, 0.664 at two), muxes less, adders and counters almost never, and a second hop adds little: bit-sliced isomorphism stalls after the first layer of logic. The control constructs are reached not because FSM logic follows the ports but because the constant compares read input ports directly (eq_const 0.614 at two hops); the FSMs themselves are not. The strict rule reaches 0.032 and 0.070, and the gate floor f3 0.914 and 0.894: at f3 a cell group lands inside a gate-level footprint almost by construction, so that figure measures the floor, not the idea. Two details of the counting, checked as exploratory recounts: a $lt, $le, $gt or $ge word with a constant port is counted as a comparator although the protocol says comparators have none (7 such words in f2; excluding them moves the datapath share from 0.0953 to 0.0948, control unchanged); and a $mux inside a mux tree and a register inside a counter are both counted (66 such constructs; deduplicating the footprints gives 0.0938, and both corrections 0.0933). Neither moves the share. The first measure we planned, a construct that is itself a port word, is empty at level 0 by construction (0 port cells in any footprint); read as a group that reproduces the construct's cells at one hop (Jaccard at least 0.5) it is 0.007 for the datapath and 0.047 for the control constructs. The honest summary is that I/O words seed registers and input-side muxes well, and nothing else.

RQ-L11: the FSM chain. Measurement only. In the f2 ground truth of VerilogEval (150 designs) 197 registers feed back to themselves, of which 16 the front end flags as FSM state; 41 constant-compare decoders read such a register, and the chains (register, decoder, select or enable) occur in 12 designs. Of the muxes with a select, 15 of 43 (VerilogEval) and 3 of 41 (micro) take it from such a decoder, 0.214 together, so the FSM, decode, select chain is real but covers a minority of the control muxes of these small designs. The detectors see little of it (Table XXVIII): in V2N they propose 41/197 feedback registers, 15/41 decoders, 5/30 sinks, all three elements of 5/89 chains, and 6/16 FSM instances as an fsm candidate. The chains are triples (register, decoder, sink) and reuse their elements: the 89 triples hold only 30 distinct sinks and 41 distinct decoders. Counted on distinct elements, the detectors propose 5/30 of the sinks and 15/41 of the decoders, against 5/89 complete triples.

TABLE XXIX
The round-4 corpus: designs per family and split, and designs that build on f1-k6, f2 and f3 with every equivalence check passing.

family	train	iid test	ood test	ok on all floors
adder	70	10	20	100
alu	70	10	20	100
comparator	70	10	20	100
counter	70	10	20	100
dpfsm	70	10	20	100
fsm	70	10	20	100
mux	70	10	20	100
pipeline	70	10	20	100
regfile	70	10	20	100
register	70	10	20	100
shifter	70	10	20	100
subtractor	70	10	20	100

Implication for a word-first detector. The classification cannot say that the wrong candidates are not noise, because the ground truth covers every cell and the noise class was unreachable; the exploratory noise′ share is 0.220 pooled and 0.522 for eq_const. The fragment share depends on the kind and, for the pooled figure, on one design and on the ff_en slices: ff_en, mux_bit and (less so) register fail as pieces of larger constructs, while eq_const fails as a detector problem (the local pattern is too weak, and a word-level view, the select and enable fan-out across a word, must decide whether a compare is a decoder). Because the fragments do not tile the words, grouping the present candidates is not enough, and a detector should construct the word first and score it as a unit. Concretely, we propose to find (i) register words by shared clock, enable and reset across flip-flops; (ii) datapath words by per-bit cones that are NPN-isomorphic, seeded from the ports but not limited to them (the port words reach registers and nothing past the first layer); (iii) control nets by select and enable fan-out across a word; and (iv) FSMs by a feedback strongly connected component plus the decoders on it. Related work attacks the word problem along the same lines: WordRev [9] derives word-level structures from the gate-level netlist of a circuit and lets the user specify sequences of word-level operations whose gates it then extracts, and DANA [6] recovers high-level registers by analysing the flow of data between individual flip-flops. We ran DANA (through HAL [4]) on the round-5 sets (Section V-J); WordRev was not run and no comparison with it is claimed. The next round builds this detector (L8, the language model given the ReverseProvenance record, stays deferred).

H. Round 4: patterns from our own data, big designs, and what is recoverable

Round 3 pointed to a word-first detector. Round 4 builds what such a detector needs: labelled data of our own, a dictionary of netlist patterns mined from it, a look at what the larger VTR designs add, and a statement

TABLE XXX
H12b on real designs (micros and the 33 VerilogEval test designs, f2): pooled F1 of the dictionary (V0 as written; V1 and V2 added after Stage 0) and of the better detector arm, and the number of designs and ground-truth constructs.

kind	dict V0	dict V1	dict V2	best detector	designs	GT
adder	0.000	0.012	0.008	0.208	29	69
subtractor	0.000	0.000	0.000	0.129	14	29
comparator	0.000	0.002	0.010	0.003	27	663
eq_const	0.000	0.000	0.008	0.001	25	643
mux	0.027	0.031	0.013	0.025	22	71
register	0.000	0.026	0.080	0.104	41	240
ff_srst	0.000	0.019	0.021	0.000	12	15

TABLE XXXI
GNN macro-F1 (f2, mean of 3 seeds) by corpus train size, on four test sets; last column: trained on the VerilogEval train split.

test set	36	72	144	288	576	840	VE train
iid	0.336	0.358	0.367	0.374	0.402	0.423	0.084
ood	0.244	0.308	0.294	0.306	0.325	0.346	0.098
micro	0.147	0.105	0.126	0.115	0.111	0.122	0.068
vetest	0.042	0.036	0.043	0.031	0.030	0.031	0.169

of what the flow destroys. Peter (Appendix A) asked for "patterns from our data (what about the bigger designs, do they provide other insights of patterns)". No language-model call and no new detector in loki/ were used; the prototypes live in experiments/round4. The rules were committed before the corpus was generated (commit c6c66bf, experiments/round4/EXPERIMENT.md); two variants of the dictionary (V1, V2) were added in a dated entry after Stage 0 and before any dictionary result, and they are reported next to the rule as written (V0), which alone gives the primary status. The thresholds were set knowing the round-2 and round-3 counts and the ground-truth kind counts of the real designs (which fixed the seven eligible kinds of H12b). Analyses added after the numbers were known (entry 2026-10-25-c of EXPERIMENT.md) are marked exploratory; they change no status. Where one shows that the measure of a rule is vacuous or confounded, the status is reported by the letter of the rule and called uninformative, with the reason.

RQ-L12a: a synthetic corpus. A seeded generator of twelve idiom families (muxes, counters, adders, subtractors, comparators, shifters, FSMs, register files, pipelines, registers, a datapath with an FSM controller, an ALU), each parameterised by width, depth, encoding, reset and enable style and coding style, gives 1,200 designs split by parameter region: train values only (840), the same region with unseen text (120, iid), and at least one test-only value, such as an unseen width or coding style (240, ood). Built by Odin III on f1-k6, f2 and f3, 1,200 of 1,200 pass every equivalence check on all three floors (Table XXIX); H12a is supported. In all 41 corpus designs have no labelled logic cell (constant shifts, constant-only comparators) and are excluded from the GNN train and test sets (the trainer needs a labelled logic node per train design); the same

TABLE XXXII
Slope of macro-F1 per doubling of the corpus train size, with 95% bootstrap interval over test designs.

floor	test set	designs	slope	95% CI
f2	iid	118	0.0171	[0.0053, 0.0261]
f2	ood	231	0.0174	[0.0092, 0.0234]
f2	micro	62	−0.0039	[−0.0100, 0.0031]
f2	vetest	29	−0.0026	[−0.0097, 0.0094]
f3	iid	118	0.0077	[0.0005, 0.0151]
f3	ood	231	0.0111	[0.0067, 0.0146]
f3	micro	59	−0.0022	[−0.0048, 0.0006]
f3	vetest	29	−0.0038	[−0.0071, −0.0009]

TABLE XXXIII
Big VTR designs at f3: cells, Odin III peak RSS (MB, all three floors), words with at least two cells and width ≥ 2, share of them in a repeat group, memories, buses of at least 8 bits, nets with at least 8 readers, FSM words.

design	cells	RSS	words	repeated	mem	buses	hi-fanout	fsm
blob_merge	31,778	4915.6	1,090	0.744	0	51	717	1
sha	10,629	841.7	225	0.924	0	29	99	0
stereovision0	56,208	1665.9	1,814	0.942	0	1263	182	0
stereovision1	142,932	1863.8	1,892	0.972	0	830	1430	48
stereovision2	254,419	4793.1	2,001	0.982	0	951	1724	0
stereovision3	859	75.8	145	0.834	0	9	10	1

filter also shrinks the test sets, so the GNN is scored on 118 of 120 iid and 231 of 240 ood corpus designs, 62 of 68 micros and 29 of 33 VerilogEval test designs, and trained on 810 of 840 corpus designs at the largest size (actual train sizes 36, 72, 144, 288, 570, 810); the dictionary scores the excluded designs with no candidates and no ground truth. About 0.400 of the iid test designs (48 of 120 at f2) and 0.108 of the ood ones (26 of 240) have a netlist, up to the multiset of cell labels and edges, identical to a train design of the same family (exploratory; this is the same collapse as the identifiability result of RQ-L14: different texts, same netlist).

RQ-L12b: a signature dictionary. For each cell the signature is the canonical class of the Boolean function of its largest cut and the Weisfeiler-Lehman context of radius r; the dictionary keeps, per construct kind, the signatures that occur at least three times in the corpus train split and lie in that kind at least 60% of the time; a matcher labels the test cells and joins labelled cells into words by adjacency or by a shared net. Against the round-2 detectors on the micros and the 33 held-out VerilogEval designs (f2, seven eligible kinds) the dictionary as written (V0) wins on none: H12b is not supported (won=0/7 floor=f2 sets=micro,vetest). V1 (cuts of up to six leaves) and V2 (a shared clock also joins storage cells), added after Stage 0, also win on none (won=0/7 floor=f2 sets=micro,vetest; won=0/7 floor=f2 sets=micro,vetest) and change no status. Table XXX gives the F1 per kind: on the real designs the dictionary finds almost nothing the corpus signatures were mined for. It does not even generalise inside the corpus: the iid test micro-F1 is

TABLE XXXIV
Style-recovery ceiling (classes of equivalent netlists over coding styles) on the twin set, per family and floor.

family	f1-k6	f2	f3
all	0.470	0.482	0.500
adder	0.667	0.667	0.667
alu	0.333	0.472	0.528
comparator	0.333	0.333	0.333
counter	0.333	0.333	0.333
dpfsm	0.500	0.500	0.500
fsm	0.500	0.500	0.500
mux	0.562	0.562	0.667
pipeline	0.333	0.333	0.333
regfile	0.500	0.500	0.500
register	0.333	0.333	0.333
shifter	0.625	0.625	0.625
subtractor	0.667	0.667	0.667

TABLE XXXV
MDL readability: description length of a candidate relative to the VerilogEval reference (median ratio) and the share of designs where the reference is strictly shorter.

variant	designs	median ratio	ref shorter
ref	156	1.00	0.000
llm	73	1.00	0.260
l0-f2	150	3.63	0.860
l0-f3	150	3.31	0.913
f2-none	150	7.89	0.940
f3-none	150	15.40	0.940
f2-calib	150	4.24	0.867
f3-calib	150	3.31	0.913

low (0.19 to 0.26) and H12b2 is inconclusive (R=0.3288 iid=0.1866 ood=0.0614 (iid < 0.20: ratio uninformative); V2: R=0.6429 iid=0.2576 ood=0.1656). Exploratory: restricted to the test designs whose netlist is absent from train (72 iid, 214 ood) the ratio R is 0.132 (V0), 0.622 (V1) and 0.622 (V2) against 0.329, and so on: duplicates do not explain the failure. Reported, not deciding: at f3 the dictionary wins on none of the seven kinds either (won=0/7), and on the corpus ood split the variants win won=1/7 (V0), won=4/7 (V1) and won=4/7 (V2), i.e. inside the corpus a dictionary does recover kinds. The word-level overlap q_K (Table XXXVI) is zero for V0 on every kind; the ceiling statement is Table XXXVII (the

TABLE XXXVI
Word-level overlap q_K (report only): share of ground-truth instances of kind K with at least two cells that a fragment of the arm covers to at least 0.8, detectors (N uncalibrated, C calibrated) and dictionary variants.

kind	instances	det N	det C	dict V0	dict V1	dict V2
adder	59	0.130	0.111	0.000	0.000	0.000
comparator	647	0.049	0.000	0.000	0.006	0.006
counter	8	0.750	0.250	0.000	0.000	0.125
eq_const	630	0.049	0.000	0.000	0.006	0.006
ff_en	134	0.045	0.000	0.000	0.000	0.000
ff_srst	15	0.667	0.133	0.000	0.133	0.067
fsm	4	1.000	0.000	0.000	0.000	0.000
mux	62	0.000	0.000	0.000	0.000	0.032
mux_tree	4	0.000	0.000	0.000	0.000	0.000
register	240	0.096	0.008	0.000	0.158	0.054
subtractor	20	0.100	0.100	0.000	0.000	0.000

TABLE XXXVII
Ceiling statement (report only): per kind, the Bayes ceiling of a signature classifier (share of the kind's cells in signatures where the kind has share at least 0.5; in-sample, 6 leaves; radius 0 and 3), the style-recovery ceiling of the family that makes the kind, and the minimum of radius 3 and style.

kind	family	B0 f2	B3 f2	style f2	min f2	B0 f3	B3 f3	style f3	min f3
adder	adder	0.72	0.94	0.67	0.67	0.35	0.95	0.67	0.67
comparator	comparator	0.81	1.00	0.33	0.33	0.52	1.00	0.33	0.33
counter	counter	0.58	1.00	0.33	0.33	0.08	0.99	0.33	0.33
eq_const	comparator	0.81	1.00	0.33	0.33	0.53	1.00	0.33	0.33
ff_en	register	0.87	1.00	0.33	0.33	0.43	1.00	0.33	0.33
ff_srst	register	0.83	1.00	0.33	0.33	0.30	0.99	0.33	0.33
fsm	fsm	0.72	1.00	0.50	0.50	0.11	1.00	0.50	0.50
mux	mux	0.83	1.00	0.56	0.56	0.47	1.00	0.67	0.67
mux_tree	mux	0.63	1.00	0.56	0.56	0.11	0.98	0.67	0.67
register	register	0.90	1.00	0.33	0.33	0.55	1.00	0.33	0.33
subtractor	subtractor	0.56	0.95	0.67	0.67	0.30	0.99	0.67	0.67

Bayes ceiling is in-sample: train and test pooled, so large radii memorise). Signatures by cone class alone are mostly floor-specific (l12b_stability.csv): few classes that fire for a kind at f2 also fire at f3.

RQ-L12c: GNN scaling. Trained on corpus subsets of 36 to 840 designs (three seeds each), the GNN improves on the corpus test sets but not on real designs (Tables XXXI, XXXII): on the 29 VerilogEval held-out designs with a labelled logic cell the slope is −0.0026 per doubling (95% interval [−0.0097, 0.0094]), so H12c is not supported. The upper bound is 0.0006 under the limit of 0.010 of the rule (not a wide margin). Exploratory: on the actual train sizes (36, 72, 144, 288, 570, 810) the slope is −0.0027 (interval [−0.0098, 0.0095]); with 20,000 bootstrap draws instead of 2,000 the upper bound is 0.0096 (nominal sizes) and 0.0098 (actual sizes), margin 0.0002, so the status holds but is close to the boundary. On the corpus iid test the slope is 0.0173 and 0.0232 when restricted to designs absent from train; on ood 0.0175 and 0.0168: the corpus gain is not a duplicate effect. A GNN trained on the VerilogEval train split itself reaches a higher macro-F1 on its held-out designs than any corpus size.

RQ-L13: big designs. Table XXXIII lists the VTR designs built one at a time with Odin III's peak resident set measured. Big designs are not equivalence-checked (agent default, as in Odin III itself; meta.json records skipped:big). 13 designs built. Odin III's reader

refuses the f3 floor (unresolved_module) of mkPktMerge, spree, boundtop, raygentop, or1200, mkSMAdapter4B, mkDelayWorker32B, whose f1-k6 and f2 floors are built (the mkPktMerge case among them); bgm, LU8PEEng, LU32PEEng exceeded the 14 GiB address limit (60% of the VM) and are skipped, partly or wholly. arm_core was stopped by the controller after 3.6 wall-clock hours (3.56 CPU hours, a hang in the linked ABC is documented for this design in the Odin III notes) without finishing its f1-k6 floor, and mcml and LU64PEEng (which Odin III's goldens do not cover) were not run; the results use the designs of Table XXXIII. Scope of the big set: the f3 big set is six designs, three of them near-clone stereovision designs (stereovision0–2: 5707 of the 7167 eligible words), no library-block design (all seven are excluded at f3 by unresolved_module) and none with a memory, so H13 says what repeats in these designs, not in big designs in general. Share of words in a repeat group in the small designs: 0.688 (1,055 designs at f3, among them 840 synthetic corpus train designs). H13a is supported (share_rep(big)=0.9283 (6653/7167) designs=6), H13b supported (big=0.9283 small=0.6879 (6233/9061) diff=0.2404) and H13c supported for the primary dictionary variant V0 (same_signature=0.8606 groups=911; V1 supported, V2 supported).

Exploratory checks of RQ-L13. (1) The repeat key has no cross-word context, so same-type same-width words with no connection to anything else could repeat trivially. Against a null in which every word is replaced by random cells with the same cell labels, the share is 0.928 observed and 0.209 null for the big designs (excess 0.720), 0.688 and 0.426 for the small designs (excess 0.262), and for the real small designs (micros and VerilogEval) 0.875 and 0.830 (excess only 0.045; VerilogEval alone 0.919 against 0.895). With a context key (the footprint cells' full-graph WL colours also must agree) the big share is 0.913 against a null of 0.034. So the repetition of the big designs is real structure (the statuses of H13a stand), while on the small real designs the key is close to its null and is not evidence. (2) The registered small baseline pools 840 synthetic corpus designs; the difference big minus small is +0.240 (supported) as registered, +0.053 against the real small designs (micros and VerilogEval: inconclusive by the rule) and +0.009 against VerilogEval alone (not supported): the status of H13b flips with the baseline, and the null-adjusted contrast (0.720 against 0.262 and 0.045) is the fairer one. (3) Only 0.020 of the big designs' words lie in repeat groups that span two or more module paths, so the repeats are mostly within a module, not module-level instance evidence. (4) For H13c the null (random label-matched footprints of the same group sizes) shares equal signatures in 0.110 of groups against the observed share of H13c (V0), so H13c is informative.

RQ-L14: what is recoverable. On the twin set (the same design in every coding style) the style-recovery ceiling is 0.470 at f1-k6, 0.482 at f2 and 0.500 at f3 (Table XXXIV): H14a is supported and H14b inconclusive (f3-f2=0.0177). The MDL score ranks the reference above the recovered RTL in the shares of Table XXXV (H14c inconclusive); its correlation with the round-2 AST similarity is rho=-0.5110 (p=1.1e-35) partial=0.3029 (p=2.1e-12) n=516 (H14d inconclusive), and the partial correlation controlling for size does not support a readability reading: the score is largely a size measure. The H14d p value pools four variants of the same 129 designs; per variant rho runs from −0.369 to −0.583, and the design-clustered bootstrap interval of the pooled rho is [-0.6183, -0.3748] (partial [0.1474, 0.4337]), exploratory. The same for CFG similarity (report only, inconclusive (report only)): rho=-0.2769 (p=1.7e-11) partial=-0.3616 (p=4.4e-19) n=571. Exploratory robustness of the identifiability result: enumerating all tied LUT input orders (720 instead of 24), using isomorphism-invariant pin labels (the registered form anonymises LUT inputs but names gate pins, so f2 could merge and f3 split netlists) or both leaves the pooled ceilings at fsIdentCapPinsOne,

fsIdentCapPinsTwo and fsIdentCapPinsThree (f3 minus f2 fsIdentCapPinsDiff), identical to the registered values; removing the 7 twin designs whose text is byte-identical to a sibling style changes them to fsIdentDedupeOne, fsIdentDedupeTwo and fsIdentDedupeThree (difference

fsIdentDedupeDiff). Exact isomorphism confirms the WL classes for fsExactThree of

fsExactThreeOf designs at f3 (fsExactTwo f2, fsExactOne f1-k6, netlists of at most 100 nodes) with fsExactDisagree disagreements with WL in fsExactBases bases. The Bayes ceiling is in-sample.

What it means for the round-5 word-first detector. The data say what not to do: this signature dictionary and this GNN did not transfer to real logic (H12b for the dictionary, H12c for the GNN), and the dictionary also fails inside the corpus (iid micro-F1 0.19 to 0.26), so the cause is not settled by the data; the corpus is a test bed for word-grouping rules, not a source of a lookup table. The detector should build words from structure that is stable across designs: registers by shared clock, enable and reset; datapath words by per-bit isomorphic cones; control nets by select and enable fan-out; FSMs by feedback components and their decoders, with the eq_const decoder decided at word level. The repeat measures of Table XXXIII show real repeated structure in the big designs (0.928 of words against a null of 0.209) but mostly within a module and near the null on small real designs, so repetition is a hint for word grouping in large designs only.

I. Round 5: the word-first detector

Rounds 3 and 4 said what to build: words from structure, the corpus as a test bed for grouping rules, repetition only where it is real. Round 5 builds it (question L Q2.4 of the purpose page: can words, control nets and FSMs be found first, and the rest built around

TABLE XXXVIII
Round 5: the six hypotheses, their status under the Stage-0 rules and the numbers the rules use.

Hypothesis	Status	Numbers
H16a	not supported	dq 0.001 q_WB 0.023 q_base 0.022 q_null 0.203 p 1.0000
H16b	inconclusive	dR 0.138 dF1 0.025 p 0.0000 n 113
H16c	not supported	dR 0.000 dF1 -0.000 p 1.0000 n 25
H16d	inconclusive	gain 2 WB 3 R2N 1 discordant 2/0 p 0.2500
H16e	inconclusive	D 0.829 null 0.715 gap 0.114
H16f	not supported	dF1 -0.0033 null share 1.000; degenerate: margin unreachable (F1 bound 0.0165)

TABLE XXXIX
Round 5: word tiling q (share of GT words whose fragments cover 0.8) at f2.

Set	R2N	R2C	W	WB	null W	null WB
MI	0.212	0.019	0.223	0.237	0.042	0.048
VT	0.022	0.004	0.017	0.023	0.017	0.203
CI	0.390	0.328	0.312	0.397	0.206	0.294
CO	0.259	0.182	0.180	0.266	0.089	0.155

them?). Two opt-in detectors, wordfirst and wordgroup (loki/re/detectors/wordfirst.py, docs/user/wordfirst.md), were added to loki re; the default detector list and every earlier calibration are unchanged. wordfirst groups flip-flops by clock, enable and reset into register words (with ff_en and ff_srst words labelled by the control net), groups 2:1 mux cells by their shared select, takes the carry chains of the arith detector so that the carry net joins the bits, decides eq_const at word level (a single-minterm cone whose support lies in one register word or the primary inputs), proposes an FSM for a register word coupled through its own D cones, and grows words along isomorphic bit slices (readers and drivers) across registers, mux words and adders into bus words. wordgroup runs last and links the per-bit hypotheses to the words that contain them. The outputs are ordinary candidates with claims, so the prover, the commit step, the scorers and the Verilog writer work unchanged. Every threshold is an agent default (no fitted parameter): groups above 32 flip-flops are split by the Q-reader and D-driver signature, eq_const supports are at most 16, growth needs 0.75 of the bits and two levels.

The rules were committed first (commit ce02fa4, experiments/round5/EXPERIMENT.md); the detector was developed on the corpus train split only. Two things differ from the committed text and are disclosed here, with the review fixes that follow from them (entries 2026-10-25-b and -c of that file). First, the FSM rule of the committed text (feedback plus at least two word-level decoders) was relaxed in the code committed at 6e97d88, eleven minutes after the protocol, to a feedback-coupling test alone; the entry that records the change was committed only at aaae4d7, after the Stage-2 numbers existed. This is a late disclosure, and commit order alone does not show whether an evaluation run preceded the change. The committed rule is reported as a labelled variant (arms W1 and WB1, Table XLV); the statuses below use the rule that was

TABLE XL
Round 5, floor f2: precision/recall/F1 per kind and set for the round-2 detectors (R2N, R2C) and the word-first arms (W, WB). Pooled over the designs each arm has; GT is the first arm's count.

Set	Kind	GT	R2N	R2C	W	WB
CI	adder	49	1.00/0.14/0.25	1.00/0.14/0.25	1.00/0.14/0.25	1.00/0.14/0.25
CI	comparator	167	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
CI	counter	10	0.62/0.50/0.56	1.00/0.60/0.75	0.00/0.00/0.00	0.62/0.50/0.56
CI	eq_const	161	0.02/0.02/0.02	1.00/0.02/0.04	0.04/0.02/0.02	0.02/0.02/0.02
CI	ff_en	78	0.00/0.00/0.00	0.00/0.00/0.00	0.20/0.18/0.19	0.02/0.18/0.03
CI	ff_srst	70	0.00/0.00/0.00	0.00/0.00/0.00	0.31/0.39/0.35	0.04/0.37/0.07
CI	fsm	20	0.89/0.80/0.84	1.00/1.00/1.00	0.09/0.05/0.06	0.58/0.75/0.65
CI	mux	106	0.73/0.08/0.14	1.00/0.07/0.12	0.67/0.06/0.10	0.40/0.08/0.13
CI	mux_tree	19	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
CI	register	163	0.57/0.37/0.45	0.59/0.26/0.36	0.27/0.30/0.29	0.36/0.40/0.38
CI	subtractor	32	0.50/0.09/0.16	1.00/0.09/0.17	1.00/0.09/0.17	0.50/0.09/0.16
CO	adder	90	0.22/0.02/0.04	0.22/0.02/0.04	0.30/0.03/0.06	0.22/0.02/0.04
CO	comparator	502	0.03/0.01/0.01	1.00/0.01/0.01	0.00/0.00/0.00	0.03/0.01/0.01
CO	counter	13	1.00/0.08/0.14	1.00/0.08/0.14	0.00/0.00/0.00	1.00/0.08/0.14
CO	eq_const	491	0.01/0.01/0.01	0.33/0.01/0.02	0.01/0.01/0.01	0.01/0.01/0.01
CO	ff_en	171	0.00/0.00/0.00	0.00/0.00/0.00	0.24/0.22/0.23	0.02/0.22/0.03
CO	ff_srst	87	0.00/0.00/0.00	0.00/0.00/0.00	0.20/0.51/0.29	0.02/0.48/0.04
CO	fsm	40	0.88/0.53/0.66	1.00/0.97/0.99	0.06/0.03/0.04	0.54/0.53/0.53
CO	mux	213	0.32/0.08/0.13	0.94/0.08/0.15	0.46/0.03/0.05	0.49/0.08/0.14
CO	mux_tree	32	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
CO	register	304	0.52/0.37/0.43	0.57/0.28/0.38	0.29/0.34/0.31	0.36/0.42/0.39
CO	subtractor	38	0.17/0.03/0.05	1.00/0.03/0.05	1.00/0.03/0.05	0.20/0.03/0.05
MI	adder	62	1.00/0.13/0.23	1.00/0.11/0.20	0.75/0.13/0.23	1.00/0.13/0.24
MI	comparator	116	0.01/0.01/0.01	0.00/0.00/0.00	0.00/0.00/0.00	0.01/0.01/0.01
MI	counter	2	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	eq_const	102	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	ff_en	4	0.04/0.50/0.07	0.00/0.00/0.00	0.14/0.25/0.18	0.05/0.75/0.09
MI	ff_srst	9	0.00/0.00/0.00	0.00/0.00/0.00	0.11/0.33/0.16	0.02/0.33/0.03
MI	fsm	0	–	–	0.00/0.00/0.00	–
MI	mux	49	0.14/0.02/0.04	0.14/0.02/0.04	0.00/0.00/0.00	0.12/0.02/0.04
MI	mux_tree	3	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	register	94	0.28/0.12/0.16	0.00/0.00/0.00	0.18/0.27/0.22	0.22/0.16/0.19
MI	subtractor	27	0.50/0.07/0.13	1.00/0.07/0.14	1.00/0.04/0.07	0.50/0.07/0.13
VT	adder	7	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	comparator	547	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	counter	6	0.67/0.33/0.44	1.00/0.33/0.50	0.00/0.00/0.00	0.67/0.33/0.44
VT	eq_const	541	0.01/0.00/0.00	0.00/0.00/0.00	0.03/0.00/0.00	0.01/0.00/0.00
VT	ff_en	130	0.00/0.01/0.00	0.00/0.00/0.00	0.01/0.02/0.02	0.00/0.02/0.00
VT	ff_srst	6	0.00/0.00/0.00	0.00/0.00/0.00	0.03/0.83/0.06	0.01/0.83/0.02
VT	fsm	4	0.40/0.50/0.44	0.00/0.00/0.00	0.00/0.00/0.00	0.33/0.25/0.29
VT	mux	22	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	mux_tree	2	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	register	146	0.11/0.05/0.07	0.00/0.00/0.00	0.06/0.06/0.06	0.05/0.06/0.06
VT	subtractor	2	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00

TABLE XLI
Round 5, floor f3: precision/recall/F1 per kind and set for the round-2 detectors (R2N, R2C) and the word-first arms (W, WB). Pooled over the designs each arm has; GT is the first arm's count.

Set	Kind	GT	R2N	R2C	W	WB
MI	adder	62	0.67/0.07/0.12	0.00/0.00/0.00	0.40/0.06/0.11	0.57/0.07/0.12
MI	comparator	116	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	counter	2	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	eq_const	102	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	ff_en	4	0.00/0.00/0.00	0.00/0.00/0.00	1.00/0.25/0.40	0.10/0.25/0.14
MI	ff_srst	9	0.00/0.00/0.00	0.00/0.00/0.00	0.07/0.22/0.11	0.01/0.22/0.02
MI	fsm	0	–	–	0.00/0.00/0.00	–
MI	mux	49	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	mux_tree	3	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	register	94	0.08/0.01/0.02	0.00/0.00/0.00	0.08/0.04/0.06	0.05/0.03/0.03
MI	subtractor	28	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	adder	7	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	comparator	547	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	counter	6	1.00/0.33/0.50	1.00/0.33/0.50	0.00/0.00/0.00	1.00/0.33/0.50
VT	eq_const	541	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	ff_en	130	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	ff_srst	6	0.00/0.00/0.00	0.00/0.00/0.00	0.10/0.33/0.15	0.01/0.33/0.01
VT	fsm	4	0.33/0.25/0.29	0.00/0.00/0.00	0.14/0.25/0.18	0.50/0.50/0.50
VT	mux	22	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	mux_tree	2	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	register	146	0.25/0.01/0.03	0.00/0.00/0.00	0.32/0.04/0.07	0.15/0.02/0.04
VT	subtractor	2	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00

evaluated. Second, the shuffles of the H16f null were drawn from one random stream instead of one stream per design as the committed text says; the committed seeding was rerun and is reported with the first (Section on H16f below). The thresholds were set knowing the round-3 and round-4 numbers (H9b 0.065, 732 ff_en candidates with one correct, 5 of 89 chains, the 0.095 hop-two reach)

TABLE XLII
Round 5, floor f1-k6: precision/recall/F1 per kind and set for the round-2 detectors (R2N, R2C) and the word-first arms (W, WB). Pooled over the designs each arm has; GT is the first arm's count.

Set	Kind	GT	R2N	R2C	W	WB
MI	adder	67	0.00/0.00/0.00	–	0.00/0.00/0.00	0.00/0.00/0.00
MI	comparator	116	0.03/0.01/0.01	–	0.00/0.00/0.00	0.03/0.01/0.01
MI	counter	2	0.00/0.00/0.00	–	0.00/0.00/0.00	0.00/0.00/0.00
MI	eq_const	102	0.00/0.00/0.00	–	0.00/0.00/0.00	0.00/0.00/0.00
MI	ff_en	4	0.29/0.50/0.36	–	1.00/0.25/0.40	0.38/0.75/0.50
MI	ff_srst	9	0.00/0.00/0.00	–	0.13/0.33/0.19	0.02/0.33/0.04
MI	fsm	0	–	–	0.00/0.00/0.00	–
MI	mux	49	0.02/0.02/0.02	–	0.00/0.00/0.00	0.02/0.02/0.02
MI	mux_tree	3	0.00/0.00/0.00	–	0.00/0.00/0.00	0.00/0.00/0.00
MI	register	94	0.35/0.20/0.26	–	0.38/0.35/0.36	0.38/0.40/0.39
MI	subtractor	29	0.00/0.00/0.00	–	0.00/0.00/0.00	0.00/0.00/0.00

TABLE XLIII
Round 5, BIG set (arm W, f2): counted, matched, GT and the same after the repetition filter F.

Design	Status	s	Counted	Matched	GT	Counted F	Matched F
stereovision3	ok	0.61	38	8	204	9	0
mkPktMerge	ok	0.59	10	10	45	6	6
boundtop	ok	2.94	136	2	731	111	0
spree	ok	9.56	58	5	443	21	0
mkSMAdapter4B	ok	3.95	374	3	530	225	0
mkDelayWorker32B	ok	1.81	5	0	910	0	0
sha	ok	9.48	69	0	402	16	0
raygentop	ok	7.61	295	6	492	230	0
blob_merge	ok	88.18	142	0	1392	59	0
or1200	ok	138.33	120	0	938	28	0
stereovision0	ok	103.01	410	10	3121	278	6
stereovision1	ok	198.23	710	85	2866	599	80
stereovision2	ok	707.5	1458	1	3544	1034	0

TABLE XLIV
Round 5: wall time per run in seconds (up to eight runs in parallel).

Arm set floor	Runs	Mean	Median	Max
W CI f2	120	0.31	0.29	0.96
W CO f2	240	0.31	0.29	0.84
W MI f1-k6	68	0.41	0.37	2.37
W MI f2	68	0.61	0.25	5.41
W MI f3	65	12.30	0.26	335.96
W VT f2	33	0.58	0.27	5.82
W VT f3	33	1.47	0.34	19.34
WB CI f2	120	0.65	0.42	3.21
WB CO f2	240	0.76	0.44	9.89
WB MI f1-k6	68	0.72	0.37	16.25
WB MI f2	68	2.82	0.29	29.06
WB MI f3	63	49.75	0.26	697.06
WB VT f2	33	2.18	0.35	44.29
WB VT f3	33	13.29	0.39	339.61

TABLE XLV
Round 5, exploratory (2026-10-25-c): the FSM rule. W and WB use the evaluated rule (feedback coupling, no decoder), W1 and WB1 the Stage-0 rule (feedback plus at least two word-level decoders). Kind-level counted/matched/GT of fsm and F1 pooled over MI, VT, CI and CO at f2; chains complete and FSM instances hit on MI and VT; the last columns classify every fsm hypothesis by the ground-truth kind of its flip-flops (FSM, counter, register that is neither, none).

Arm	Rule	fsm c/m/GT	F1	Chains	FSM hit	Hyp.	FSM	Counter	Register	None
R2N	per-bit	47/39/64	0.70	1/22	2/4	–	–	–	–	–
R2C	per-bit	59/59/64	0.96	1/22	0/4	–	–	–	–	–
W	round 5	30/2/64	0.04	2/22	0/4	138	59	31	43	5
WB	round 5	68/37/64	0.56	3/22	1/4	165	62	37	62	4
W1	Stage 0	6/1/64	0.03	2/22	0/4	15	8	6	1	0
WB1	Stage 0	45/37/64	0.68	3/22	2/4	14	7	6	1	0

and the ground-truth kind counts of the evaluation sets. The evaluation sets are the micros (MI), the 33 held-out VerilogEval designs (VT), the corpus iid and ood test splits (CI, CO) and the 13 big designs that built at f2 (BIG). The arms are the round-2 detectors uncalibrated and calibrated (R2N, R2C), wordfirst alone (W) and wordfirst with the per-bit detectors and wordgroup (WB); all runs are uncalibrated. Table XLVIII lists the questions.

Outcome. Under the Stage-0 rules: H16a (tiling, VT) is not supported, H16b (ff_en words) inconclusive, H16c (word-level eq_const) not supported, H16d (chains) inconclusive, H16e (growth beyond a null) inconclusive and H16f (repetition on BIG) not supported (Table XXXVIII). No hypothesis is supported. Two of the statuses carry a qualification: H16f is degenerate (its margin cannot be reached on this data, see below), and the null clause of H16a is empty on VT.

H16a: tiling. On VT the pooled tiling of WB is 0.023 against 0.022 for the better round-2 arm (difference 0.001, bootstrap interval 0 to 0.062, 1,388 instances over 17 designs, Wilcoxon p 1). Word grouping does not tile the held-out real words; on the micros and the synthetic splits the tiling of WB and of the round-2 arm are close (Table XXXIX). The null columns replace each counted candidate by random cells of the same size. On VT the null of WB (0.203) is above WB itself and above every arm (the largest arm value is 0.023), so the null clause of the Stage-0 rule (q(WB) minus the null at least 0.10) cannot hold on VT. The Stage-0 reachability sentence (WB at least 0.215) used the round-3 baseline and ignored that clause; with the VT baseline the difference clause needs q(WB) of 0.172 and the null clause 0.303, so a supported verdict needed q(WB) of about 0.303. The status does not depend on the null: the difference is below 0.05. The null depends on the size mix of the candidates: WB has 34 counted candidates of at least ten cells on VT against 20 for R2N, the largest has 1,016 cells (R2N: 64), and the null of all of WB's counted candidates is 0.203 against 0.203 for R2N, while the ff_en candidates alone give 0.192 for WB and 0.004 for R2N (exploratory, Table XLVI). The tiling measure cannot separate structure from chance when a few candidates are large.

H16b: ff_en words. Pooled over all four sets the recall of ff_en rises from 0.008 (best round-2 arm) to 0.146 for WB (precision 0.001 to 0.014); the gain in recall (0.138) and in F1 (0.025) is below the Stage-0 margins (0.20 and 0.10) although the paired test over 113 designs is significant, so the status is inconclusive. Grouping by shared control matches some enabled words that per-bit slices never matched, at a precision of 0.014 (F1 0.026); the recall stays at 0.146. Why it stops there was not measured. Three reasons are plausible and untested: the

TABLE XLVI
Round 5, exploratory (2026-10-25-c): tiling on VT and its random null by candidate subset. Candidates of at least ten cells and the largest candidate drive the null; the same ff_en candidates give a null of a few percent for R2N and 0.19 for WB.

Arm	Subset	Candidates	≥10 cells	Max cells	q	q null
R2N	all	1453	20	64	0.022	0.203
R2N	ff_en	732	0	5	0.003	0.004
R2N	other	721	20	64	0.022	0.018
WB	all	1845	34	1016	0.023	0.203
WB	ff_en	867	1	11	0.003	0.192
WB	other	978	33	1016	0.023	0.203

TABLE XLVII
Round 5, exploratory (2026-10-25-c): arm WB on BIG with the Stage-0 cap of 1800 s per design (smallest first). Counted/matched for W and WB on the designs where WB finished; GT words of the design.

Design	WB status	s	W c/m	WB c/m	GT
stereovision3	ok	8.45	38/8	285/7	204
mkPktMerge	ok	1.0	10/10	238/18	45
boundtop	ok	27.6	136/2	798/2	731
spree	ok	40.14	58/5	502/9	443
mkSMAdapter4B	ok	175.38	374/3	1747/3	530
mkDelayWorker32B	ok	73.0	5/0	460/0	910
sha	ok	91.88	69/0	1185/0	402
raygentop	ok	111.01	295/6	1245/7	492
blob_merge	ok	848.46	142/0	2164/0	1392
or1200	ok	733.63	120/0	1675/0	938
stereovision0	ok	651.97	410/10	13090/12	3121
stereovision1	timeout	–	–	–	–
stereovision2	timeout	–	–	–	–

TABLE XLVIII
Round 5 research questions. Each serves L Q2.4; the measure is the recovery of the Odin III ground truth (precision, recall, word tiling), not area, speed or power.

RQ	Question
L16a	Do word-first candidates tile the ground-truth words (round 3: 0.065)?
L16b	Does clock, enable and reset grouping recover the enabled words?
L16c	Does word-level eq_const beat the per-bit detector?
L16d	Does the FSM, decode, control chain complete more often (round 3: 5 of 89)?
L16e	Do words grown from registers reach the datapath beyond a size-matched null?
L16f	Is repetition real evidence on big designs, against shuffled groups?

enable is often a function of several nets, the cone depth limit of the ff_pattern analysis cuts it, and the ground-truth footprints of such words are whole arrays that share cells.

H16c: word-level eq_const. The word-level decision changes nothing measurable: F1 is 0.001 precision and 0.002 recall for WB against 0.001 and 0.002 for the baseline, over 25 designs with the kind (not supported). WB says nothing about the word-level rule, and the reason is in the code. The per-bit eq_const of arith applies the same single-minterm test to every cone with at most 16 support nets; the word-level rule applies it only to cones whose support lies in one register word or the primary inputs, so it is a subset. On MI and VT (98 designs) the number of counted word-level candidates of W with cells that R2N lacks is 0, and WB ends with the same cell sets as R2N in 97 of those designs (163 against 163 counted on VT): the overlay merges two candidates of equal kind, cells and parameters into one hypothesis, so a word-level candidate that arith also proposes is not a second one and WB does not drop it. Arm W alone is the reading of the word-level rule (exploratory variant, no status): it counts 39 candidates on VT against 163 for R2N at the same recall, with F1 0.002 against 0.001. Why the word-level rule finds fewer cones is not established. One development design (a corpus FSM design in which none of the LUTs is a pure decoder) suggests that the f2 mapping absorbs decoders into next-state LUTs; this was not measured on the evaluation sets. Across the four sets only 27 of the 138 fsm hypotheses of W contain a word-level decoder (exploratory). The ne_const branch of the detector has a defect (the inverted table is masked with the wrong width, so a constant of four or more bits reads back as −1 and the prover refutes the candidate): 3 of 48 ne candidates of W on MI and VT are refuted by it. It was left in place so that the evaluated results stand.

H16d: the chain. Of the 22 ground-truth chains of MI and VT, WB completes 3, W 2 and the round-2 uncalibrated arm 1 (calibrated: 1); the two chains that WB completes and R2N does not are fewer than the margin of five and fewer than the eight discordant chains the test needs (with fewer than eight discordant chains the rule gives inconclusive whatever the gain). FSM instances matched: R2N 2 of 4, WB 1, W 0. The FSM rule that was evaluated (a feedback-coupling test, no decoder required) over-generates. Of the 138 fsm hypotheses of W on the four f2 sets, 59 have their flip-flops in a ground-truth FSM, 31 in a counter, 43 in a register that is neither (the ground truth has no shift-register or LFSR kind, so these fall here) and 5 in none; of the 30 that are counted only 4 are FSMs and 8 are counters (exploratory, Table XLV). The committed Stage-0 rule (W1, WB1) proposes 15 and 14 hypotheses in W and WB where the evaluated rule proposes 138 and 165; the chains it completes are 2 (W) and 3 (WB), the FSM instances it hits 0 and 2, and the pooled F1 of kind fsm is 0.679 for WB1 against 0.561 for WB (R2N 0.703, R2C 0.959). Of the 50 W hypotheses that match a ground-truth FSM (Jaccard at least 0.5) only 2 are counted, and 48 conflict with a register hypothesis over the same cells and stay open below $p = 0.5$. Which features keep them below that was not measured.

H16e: growth. The seeded share of the datapath constructs is 0.829 for W (WB 0.788, R2N 0.754), far above the 0.095 of I/O-port propagation, but a size-matched random null reaches 0.715: the gap (D 0.829 null 0.715 gap 0.114) is below the margin and the level alone is not

evidence, because the ground-truth instances partition the non-I/O cells. By kind, the gap to the null is smallest for registers (0.078: 0.921 against 0.843 over 240 constructs), the kind the growth starts from, and larger for subtractors (0.129), adders (0.176), comparators (0.190) and muxes (0.219); counters and mux trees have a gap of zero (8 and 4 constructs, both at the ceiling of their null). The per-kind numbers come from r5_numbers.csv; the kinds are few and no interval was computed for them.

H16f: repetition. On the 13 BIG designs W counts 3,825 candidates and matches 130 of 15,618 ground-truth words; keeping only the candidates in a repeat group lowers the matches to 92 and F1 from 0.013 to 0.010, and every shuffled draw reaches that F1 or more (share 1): repetition of the candidates themselves is not evidence of a correct word here, in line with the near-null repeat key on small designs in round 4. H16f is degenerate: even a filter that kept exactly the matched candidates would reach an F1 of 0.017, a gain of at most 0.003 over keeping all, below the margin of 0.05, so no outcome but not supported was reachable and the literal status says nothing about repetition. The first run drew the shuffles from one random stream, which Stage 0 does not say; with the Stage-0 per-design seeding the share of null draws at or above F1 of the filtered set is still 1 (null mean 0.011, against 0.011 for the stream), and the status is unchanged. Arm WB was run on BIG as an exploratory addition with the Stage-0 cap of 1800 s: it finished 11 of 13 designs (2 timed out; Table XLVII). On those designs W counts 1,657 candidates and matches 44 of 9,208 words (F1 0.008, 0.002 with the repetition filter); WB counts 23,389 and matches 58 (F1 0.004, 0.001 filtered). The filter lowers F1 for both arms.

Run time. wordfirst is cheap: the median W run takes 0.25 to 0.29 s per design on MI, VT, CI and CO (by set), and W takes between 0.6 s (mkPktMerge) and 708 s (stereovision2) on BIG with –features none; the median WB run takes 0.29 to 0.44 s on the same sets (Table XLIV, Table XLIII). WB hit the 900 s limit on 2 micro f3 designs (f3 is report-only).

What remains for L15. The loss audit (L Q3.1) asks which pass loses each construct. The results above separate two reasons a word is not found. For registers and enable words the information is in the netlist and the detector or the grouping rule misses it (a fixable detector problem). For FSM decoders, comparators and the word boundaries of muxes the evaluated rules found nothing beyond the per-bit detectors (H16c) and the data do not show why. One development design suggests that the f2 mapping absorbs decoders into next-state LUTs, and register-file words are known to share one ground-truth footprint; neither is measured on the evaluation sets, and whether any word-level rule could recover these constructs is open. Which pass erases them (lowering of the $eq, the absorption into LUTs, the merging of mux and enable) needs the construct chain that the Odin III D14 provenance export will give at every granularity; with it the audit can label each missing construct with the pass that joined it to its neighbours.

TABLE XLIX
The pre-registered decision on E_real at f2 under the two scorers for Loki (HAL counts identical in both rows). $D = F_J(\text{Loki}) - F_J(\text{DANA})$; p_L, p_H: one-sided Wilcoxon p, Loki greater and DANA greater. The last column applies the rule to that row alone; the pre-registered outcome needs Loki ahead under both.

Scorer	D	95% CI	90% CI	p_L	p_H
Comparison scorer	-0.092	[-0.317, 0.043]	[-0.266, 0.009]	0.828	0.172
Round-5 scorer	-0.132	[-0.382, -0.044]	[-0.337, -0.052]	0.999	0.001

Limits. The detector reuses the ff_pattern cone analysis (depth three, twelve inputs) and the arith chain finder, so its adders are those of the baseline; carry chains are not rebuilt from isomorphic slices. At f3 every arm has zero recall on muxes and comparators, and the word-level eq_const of W counts 898 candidates on MI and VT for 1 match (R2N: 3,405 for 1; Table XLI). The rules were developed on 150 synthetic train designs and did not transfer to the real held-out set, the same pattern as the dictionary of round 4. The FSM test over-generates (counters, LFSRs and other coupled registers pass it) and the ne_const branch has the constant-mask defect described under H16c.

J. Round 5b: the word-first detector against HAL's DANA

Question L Q2.4 asks whether words can be found first; this subsection asks whether an established word finder does better on the same netlists. The comparison was pre-registered (docs/plan/hal-compare-prereg.md, commit 363187f) before HAL was built, with two dated amendments: Amendment 1 (commit 4a71877, before any run; the scorer differences, the bootstrap interval, the equivalence rule) and Amendment 2 (commit 373049e; the HAL route and the results of the load checks). Amendment 2 was written after a HAL load of the 100 MI and VT f2 designs that compared only the LUT truth-table bit order of our library with the BLIF covers; that check computed no DANA grouping and no word-recovery score (erratum in the pre-registration). The tool is HAL v4.5.0 (commit e7fea566b51e) with its dataflow plugin DANA [4], [6], built without the graphical interface. Both tools read the same netlist.blif of a floor; a converter turns it into structural Verilog of the cells of one of two gate libraries that we wrote for HAL (a LUT library for f2, the cmos6 library for f3), with the instance name equal to the Odin III node id so that a DANA group is a set of answer-key nodes. The HAL v4.5.0 command line –dataflow does not run (its handler sets no gate types), so DANA was driven through the plugin's Python API with its documented default configuration; only min_group_size was set. The library default of min_group_size (8) is the headline arm; 2 is a sensitivity run. Loki's side is the round-5 W arm, not rerun. Ground truth are the

TABLE L
Loki word-first detector (W) against HAL's DANA on MI and VT (E_real), word recovery against the L0 ground truth; one scorer for both tools. P_J, R_J, F_J: Jaccard-0.5 matching; purity and completeness over flip-flops. The round-5 scorer row counts Loki as round 5 did (Amendment A1). Loki WB has a run on fewer designs than the other arms (f3: 96 of 98 designs; listed in docs/results-hal.md), so its rows pool a different design set.

	f2 (LUT6)					f3 (cmos6)				
Arm	P_J	R_J	F_J	pur.	compl.	P_J	R_J	F_J	pur.	compl.
Loki W	0.160	0.179	0.169	0.907	0.063	0.557	0.142	0.226	0.952	0.850
Loki W, round-5 scorer	0.119	0.142	0.129	-	-	0.149	0.042	0.065	-	-
Loki WB	0.141	0.133	0.137	0.983	0.129	0.320	0.071	0.116	0.956	0.874
DANA, size 8 (headline)	0.745	0.158	0.261	0.853	0.550	0.745	0.158	0.261	0.853	0.550
DANA, size 2	0.613	0.204	0.306	0.865	0.549	0.613	0.204	0.306	0.865	0.549

L0 storage words with WIDTH ≥ 2 whose footprint, cut to flip-flops, holds at least two flip-flops, scored by one code path for both tools (Jaccard-0.5 one-to-one matching, exact match, purity, completeness).

Table L gives the pooled result on MI and VT. On f2 the pooled F_J of DANA is 0.261 against 0.169 for Loki's W arm over 100 designs (41 of them with a ground-truth word); the difference $D = F_J(\text{Loki}) - F_J(\text{DANA})$ is -0.092, with a 95% cluster-bootstrap interval of [-0.317, 0.043]. The pre-registered rule returns inconclusive: the point estimate favours DANA, but the interval is wide (the micro designs come in template families), the paired Wilcoxon test over the designs with a ground-truth word does not reach $p < 0.05$ (0.172 for DANA greater), and the interval is not inside the equivalence band either, so neither a lead nor a tie is shown. The comparison scorer (duplicate predictions counted once, Loki's cells and the ground-truth footprints cut to flip-flops) was committed (ca4cbac, 2026-10-26 17:37) after Loki's round-5 counts were known (round 5, 2026-10-25) and before any DANA grouping was scored; it moves Loki's pooled f2 F_J from 0.129 to 0.169. The decision is therefore shown under both scorers (Table XLIX; DANA's counts are the same in both rows): D is -0.092 with the comparison scorer (95% interval [-0.317, 0.043]) and -0.132 with the round-5 scorer ([-0.382, -0.044]); the rule applied to each alone gives inconclusive and inconclusive, and neither scorer is the headline. DANA's size-2 run gives a pooled F_J of 0.306; it is a sensitivity run, never the headline. Its pre-registered outcome is also not a decided one, but with the round-5 scorer alone the rule gives HAL ahead for it (comparison scorer alone: inconclusive).

The two tools differ in kind. DANA proposes fewer groups (51 against 268 for Loki on f2); a larger share of them matches a word (0.745 against 0.160), but few words are recovered by either tool (recall 0.158 for DANA, 0.179 for Loki). DANA's groups cover most flip-flops of the ground-truth words (0.985 against 0.843) and are more complete (0.550 against 0.063), with a lower purity (0.853 against 0.907). Its groups are larger (median 10.0, mean 37.3 flip-flops, against 2.0 and 6.0 for Loki on f2), and 0.25 of them (13) merge words, that is, hold flip-flops of a ground-truth word and of another one, against 0.07 (18) for Loki. Loki's groups are purer but split words and leave flip-flops out. On f3 DANA's rows equal its f2 rows: it reads the flip-flop graph and control nets only, and the f2 and f3 netlists of a design have the same flip-flops and the same flip-flop connectivity; DANA's partitions are equal on 98 of 98 designs (under the original flip-flop names) and no netlist holds a DFFSR cell (0 in 199). Loki's f3 numbers differ from f2 (Table L); at f3 Loki's precision is higher and its recall lower. This shift comes mostly from the scorer's flip-flop cuts (Loki's cells and the ground-truth footprints), which raise the match count only together, and not from the detector: with the round-5 scorer Loki's pooled f3 F_J is 0.065 (precision 0.149, recall 0.042), against 0.226 with the comparison scorer (Amendment 1 table in the pre-registration).

Limits. The 13 large designs were not scored: 7 contain a RAM hard block that neither gate library can express, which is above the pre-registered 20% drop limit; DANA ran on the 6 that convert (at most 46 s each) and the group counts are in docs/results-hal.md. No evaluation netlist contains a flip-flop with set and reset pins, so DANA's control-net evidence is the clock only on both floors; a netlist with enable and reset flip-flops is the case DANA was designed for, and we did not test it. The converter was checked against the BLIF by simulation (micro, VerilogEval and the convertible large designs); the 10545 LUT truth tables as HAL parses them show 0 mismatches with the BLIF, and the BLIF and HAL LUT counts differ on 0 of 100 designs; DANA returned the same groups on a second run (5 of 5 designs, and 5 of 5 with both runs uncached) and with permuted instance names and shuffled instance order (6 of 6 designs); these two checks cover only small VT designs (the 5 smallest designs with a ground-truth word, and 6), no MI design; HAL was not patched and DANA was not tuned. The ground truth is the Odin III answer key, so a group that matches no word is wrong only in that sense. For L Q2.4 the result is limited to register words on these sets: under the pre-registered rule the word-first rules are not shown to be ahead of, or equal to, a published dataflow method, and neither tool recovers most of the words.

TABLE LI
LLM pilot, run re: pass rate per arm (strict scorer unless marked). $n = 40$ problems per arm, one draw each, claude-sonnet-5-5. Arm A: netlist; B: netlist plus Loki hints; C: interface only. GB-equiv.: Gate-Breaker-equivalent scorer.

View	A	B	C
All 40, refusal/no file = fail (strict)	35/40 (0.88)	35/40 (0.88)	30/40 (0.75)
All 40, refusal/no file = fail (GB-equiv.)	33/40 (0.82)	33/40 (0.82)	29/40 (0.72)
Answered only (strict)	35/40 (0.88)	35/38 (0.92)	30/40 (0.75)
Paired set, 38 (strict)	33/38 (0.87)	35/38 (0.92)	29/38 (0.76)
Paired set, 38 (GB-equiv.)	31/38 (0.82)	33/38 (0.87)	28/38 (0.74)

TABLE LII
LLM pilot, run re: exact two-sided McNemar tests, paired set $n = 38$, strict scorer. "First only" counts problems the first arm passes and the second fails.

Comparison	Both	First only	Second only	Neither	p
B vs A	33	2	0	3	0.5000
A vs C	28	5	1	4	0.2188
B vs C	28	7	1	2	0.0703

TABLE LIII
LLM pilot: API refusals and no-file calls per arm. standard stopped at order 88 of 120 (calls made); re ran all 120.

Run	Arm	Calls	API refused	No file	Total
standard	A	30	2	4	6/30 (0.20)
standard	B	29	3	1	4/29 (0.14)
standard	C	29	3	2	5/29 (0.17)
re	A	40	0	0	0/40 (0.00)
re	B	40	0	2	2/40 (0.05)
re	C	40	0	0	0/40 (0.00)

K. LLM pilot: do Loki hints help a model recover RTL?

Every other result in this paper uses no language model. This subsection reports one small pilot that does, for research question L Q1.2: does an LLM recover RTL from a flat gate netlist better when it is given Loki's structural hints? The measure is the functional pass rate per arm, plus the refusal rate.

Setup. The pilot is a replica of the Gate-Breaker netlist-to-RTL task [1] on VerilogEval [2]. It uses 40 problems, 10 per complexity quartile, chosen by a fixed hash rank from the quartile files, and one call per problem per arm ($K = 1$). There are three arms. Arm A gives the gate netlist and the interface line. Arm B adds the claims that Loki makes from the flat netlist alone (no provenance names, no L0 ground truth). Arm C gives the interface line only and no netlist, as a memorisation control. All arms use the same two worked examples and the same answer format. Each call is a fresh claude-sonnet-5-5 subagent with no prior turns. The strict scorer compiles the reply with Icarus Verilog 12.0 against the reference testbench and requires exactly one line Mismatches: 0 in M samples; it also rejects replies that name the hidden reference module or print their own result line. A Gate-Breaker-equivalent count is reported beside it as a cross-check. A call with no reply file or an API refusal counts as a fail in the all-calls view.

Results. Table LI gives the pass rate per arm in both views, and Table LII gives the exact two-sided McNemar tests on the paired set of 38 problems answered in all three arms. Arm B is nominally ahead of arm A in the paired view (35/38 against 33/38) and both are ahead of arm C (29/38), but no difference reaches $p < 0.05$: the smallest is B against C at $p = 0.0703$. This is a pilot at $n = 40$ with one draw per cell; it cannot support a claim about absolute rates or about the effect of the hints. In the all-calls view arms A and B are equal (35/40) because arm B's two missing reply files count as fails, and arm A passed both of those problems. Per-quartile cells have 8 to 10 problems each; in Q4 the all-calls rates are A 9/10, B 7/10, C 7/10, so the sign of the B-minus-A difference is not stable across quartiles.

Refusals. The first complete prompt that we ran, the Gate-Breaker standard prompt, was stopped by the pre-set rule at 88 of 120 calls: 8 calls came back as API safeguard refusals (no model reply) and 7 as no-file calls (the subagent declined or answered in chat without writing the reply file). Under our own prompt re, which explains the research, the same 120 calls gave 0 API refusals and 2 no-file calls. Table LIII gives the rates per arm. The two prompts differ in more than the refusal behaviour, so any comparison of pass rates between them is exploratory only. Refusal handling is part of the cost of using a general-purpose model for this task and has to be reported beside the pass rate.

Threats to validity. (i) The subagents were not tool-less. They had file and web tools loaded and the workspace instructions and a memory index in context, and the reference and test files sit on the same disk (decision 11, a blinding deviation that was accepted). The transcript audit of all 208 calls (standard 88, re 120) shows that no call used any tool other than reading its own call file and writing or editing its own reply file: no shell, search or web tool. This is evidence from behaviour, not a technical block. Two calls used more than the usual two tool calls (standard 073 and re 116); both touched only their own files and both are scored. (ii) Decision 12: the first batch used a step-by-step reasoning prompt (cot) that the API safeguard refused in 6 of 10 calls, even in arm C; those answers were discarded and the prompt replaced by standard. Decision 13: a refusal is never retried or reworded, and pass rates are reported with refusal as a fail and on the paired set. Decision 14: the re system prompt is ours, not Gate-Breaker's, so exact Gate-Breaker fidelity is dropped; the worked examples, user content and answer format are unchanged (120 of 120 user messages byte-equal). The re prompt mentions Loki hints in every arm, including arms A and C that carry none; this cue is the same in all arms. (iii) The prompts therefore differ from

TABLE LIV
Round 6: units per set at the stmt level and what became of them.

Set	Designs	Units	Kept	Lost	Born joined	Capped
ALL	221	1436	474	580	382	21
CI	120	765	179	304	282	20
MI	68	546	253	193	100	1
VT	33	125	42	83	0	0

TABLE LV
Round 6: first losses of a unit per hardware kind and pass, real/null (stmt level, MI+VT+CI). The first column is the units at risk at opt; the null is the expected loss of the random-set null.

Kind	n	fsm	opt	flatten	opt -full	wreduce	lower	abc
fsm	0	–	–	–	–	–	–	
register	218	84/94	0/35	4/35	0/31	0/7	1/13	
arith	172	17/51	0/36	7/36	0/35	0/0	31/22	
mux	359	192/223	0/57	14/61	0/45	0/10	79/36	
compare	33	18/14	0/3	0/1	0/2	0/1	1/1	
logic	272	9/81	68/58	1/38	0/39	0/13	54/46	

Gate-Breaker's. Sampling is one draw with no controlled temperature, and the 40-problem subset has a prompt-size cap that under-represents big designs, so these rates are not comparable with the published ones. The hints are uncalibrated at f3, and arm C controls for memorisation of VerilogEval; it is not a measure of recovery.

L. Round 6: the L15 loss audit

Round 5 found that register and enable words are findable and that FSM decoders, compares and mux boundaries are not, and guessed that they are absorbed into LUTs at f2. Round 6 asks where and why (question L Q3.1 of the purpose page: for each HDL construct, at which Odin III step is it first lost as a separable unit, joined with what, and why?). The measure is a count of constructs lost per kind and pass, the share that the D14 provenance explains, and the recall of the round-5 detectors before and after a pass. It is not area, speed or power. The tools are Odin III [10] at c82609e6, which writes the version 2 name map (–prov-map: records, join labels and the construct table), and ABC [11] inside its abc pass; no VPR run is involved.

Protocol. The rules were committed first (commit 6d0e5b5, experiments/round6/EXPERIMENT.md) and the audit was run on no evaluation design before it. Development used hand-written fixtures and the corpus train split. The f2 flow of round 5 is run pass by pass in one Python session (proc, fsm, opt, flatten, opt -full, wreduce, lower, abc) and a netlist plus map is written after each pass. A unit is the statement-level anchor of a record's construct leaves (V1.2.0 Section 3.3), reduced by dropping ancestors; a unit is separable at a snapshot when some logic object has exactly that unit set, and lost at the first later snapshot where none has. A loss is explained when a labelled same-pass join record names the unit; its reason class is read from the pass and that evidence. The null draws, for every object, a random unit

TABLE LVI
Round 6: first losses per reason class and pass (stmt level, MI+VT+CI). Classes with no loss are kept.

Reason	fsm	opt	flatten	opt -full	wreduce	lower	abc
removed	12	68	7	0	0	10	
subsumed	127	0	1	0	0	33	
fsm-recode	0	0	0	0	0	0	
instance-join	0	0	0	0	0	2	
opt-merge	77	0	17	0	0	0	
width-reduce	0	0	0	0	0	0	
lower-share	0	0	0	0	0	0	
lut-cone	0	0	0	0	0	99	
unlabelled-merge	104	0	1	0	0	22	

TABLE LVII
Round 6: merged first losses, those a labelled same-pass join record names (explained), the unlabelled rest and how many of those are inherited blends (exploratory). stmt level, MI+VT+CI.

Pass	Merged	Explained	Share	Unlabelled	Inherited
S3_opt	181	77	0.425	104	104
S5_opt_full	18	17	0.944	1	1
S8_abc	123	101	0.821	22	22
all	322	195	0.606	127	127

set of the same size from the units alive at the pass, so that growth of the chains is matched. Four dated entries follow in the protocol file. Two were written on development designs before any evaluation run: removed is split into removed and subsumed (a statement that survives only as the enclosing statement of a finer unit), and the units that become separable after the first snapshot are counted as losses with a flag. Two are post-data and exploratory: an axis that moves the instance cells to a class of their own, and the count of inherited blends below. The numeric rules use the original axis and change no status. The sets are the 68 micros (MI), the 33 held-out VerilogEval designs (VT) and the 120 corpus iid designs (CI), each with all designs ok; the 13 big designs and the ood split were not run. The snapshot writers do not change the netlist (the final BLIF of all 68 micros is byte-identical to a run without them).

Where units are lost. Of 1436 units, 580 are lost, 474 are never lost and 382 are joined from the first snapshot onwards (Table LIV). The losses fall on four passes only: 320 at opt, 68 at flatten, 26 at the second opt and 166 at abc. None is lost at fsm, wreduce or lower: lower splits word cells into gates, which makes units finer, not blended. The 68 at flatten are all INSTANCE units (the instance cell of a dissolved module) and say nothing about logic. By hardware kind (Table LV), registers (dominant) and compares (dominant) are lost at opt (84 of 89 and 18 of 19, against random-set null shares of 0.34 and 0.49), muxes at opt (192 of 285, null share 0.41) and arithmetic at abc (31 of 55, null share 0.10). By the count rule, abc loses muxes more than chance (79 against 36) and arithmetic at chance, whereas arithmetic at opt (17 against 51) and muxes at the second opt (14 against 61) are below chance. The round-5 reading is therefore only half right: a compare statement is already not a separable unit after the first

TABLE LVIII
Round 6, rule 1: the pass that loses most of each hardware kind, its share and the null share (stmt, MI+VT+CI).

Kind	Pass	Losses	Share/null	Outcome
arith	S8_abc	55	0.564/0.101	dominant
compare	S3_opt	19	0.947/0.489	dominant
logic	S4_flatten	132	0.515/0.168	dominant
mux	S3_opt	285	0.674/0.412	dominant
register	S3_opt	89	0.944/0.343	dominant

TABLE LIX
Round 6: recall per kind of the three arms on the gate netlist (S7) and the LUT netlist (S8), MI+VT+CI pooled; GT is the S7 count.

Kind	GT	B S7	B S8	W S7	W S8	WB S7	WB S8
ff_en	212	0.00	0.01	0.16	0.11	0.16	0.12
ff_srst	85	0.00	0.00	0.53	0.49	0.53	0.47
register	397	0.13	0.20	0.22	0.23	0.22	0.23
adder	110	0.00	0.15	0.00	0.14	0.00	0.14
subtractor	63	0.00	0.08	0.00	0.05	0.00	0.07
comparator	829	0.02	0.00	0.00	0.00	0.02	0.00
eq_const	803	0.00	0.01	0.00	0.00	0.00	0.01
mux	174	0.53	0.06	0.36	0.04	0.55	0.07
mux_tree	24	0.04	0.00	0.00	0.00	0.04	0.00
counter	18	0.00	0.39	0.00	0.00	0.00	0.39
fsm	24	0.12	0.75	0.04	0.04	0.17	0.67

opt, before any LUT is formed, and the LUT cone (abc) is where arithmetic and the remaining mux statements go. The fsm pass leaves no mark: there is no unit of hardware kind fsm at the first snapshot, and the FSM-recode reason class has no observation. The statement's branch level gives the same picture (548 losses against 580; l15_kind_pass.csv).

Why. Table LVI gives the reason classes. At opt, 77 losses are opt merges that a labelled join record names, 12 are removals, 127 are subsumptions and 104 are merges with no label; at abc, 99 are LUT-cone merges that a record names, 33 subsumptions, 10 removals, 2 instance joins and 22 unlabelled merges. Of the 322 merged losses, 195 (0.606) are explained by a same-pass join record, which the Stage-0 rule reads as partly explained (Table LVII). Every one of the 127 unlabelled merges is an inherited blend (exploratory): the object that remains had already blended the unit with others before the pass, so no new join was made and none is labelled (V1.2.0 Section 3.3). The D14 labels therefore account for every new blend; what they do not say is that a pure object was absorbed into an existing blend. "Style collapse" (a case, an if and a ternary of one function) leaves no record and is reported as not observable. Table LX shows eight lost micro constructs with their partner; l15_lost_list.csv holds all of them.

Detectors. The word-level snapshots are not netlists a detector reads, so recall is compared at the two netlist snapshots, the gates after lower (S7) and the LUTs after abc (S8), for the default list (B), wordfirst (W) and the combined list (WB). The ground truth is the round-5 derivation on the word cells of S6 with the span-

TABLE LX
Round 6: eight lost micro constructs, 'construct at file:line lost at pass P because merged with Q' (first rows of l15_lost_list.csv with one labelled partner).

Design	Construct	Kind	Pass	Joined with
bm_expr_all_mod	bm_expr_all_mod.v:124.2-124.29	NONBLOCKING_ASSIGN	opt	NONBLOCKING_ASSIGN 146.2-146.122
bm_expr_all_mod	bm_expr_all_mod.v:133.2-133.39	NONBLOCKING_ASSIGN	opt	NONBLOCKING_ASSIGN 147.2-147.126
bm_lpm_all	bm_lpm_all.v:44.8-44.20	NET_ASSIGN	opt	NET_ASSIGN 51.8-51.43
bm_match1_str_arch	bm_match1_str_arch.v:49.8-49.26	NET_ASSIGN	opt	NONBLOCKING_ASSIGN 55.2-55.22
bm_match1_str_arch	bm_match1_str_arch.v:50.8-50.26	NET_ASSIGN	opt	NONBLOCKING_ASSIGN 56.2-56.22
bm_match1_str_arch	bm_match1_str_arch.v:51.8-51.26	NET_ASSIGN	opt	NONBLOCKING_ASSIGN 57.2-57.22
bm_match1_str_arch	bm_match1_str_arch.v:55.2-55.22	NONBLOCKING_ASSIGN	opt	NET_ASSIGN 49.8-49.26
bm_match1_str_arch	bm_match1_str_arch.v:56.2-56.22	NONBLOCKING_ASSIGN	opt	NET_ASSIGN 50.8-50.26

subset footprint rule of L0 applied to construct keys; at S7 many gates share a construct set, so S7 footprints are broad and the S7 recall is a lower bound, which is why the table is read through rule 4 only. One design (ve_Prob021_mux256to1v) was killed at S7 by the 6 GB slice cap and two micro WB runs at S7 timed out; they drop out of the paired comparison. Under rule 4 the only kind that abc fools is the mux word, in all three arms (WB: recall 0.55 to 0.07); the FSM and counter kinds are found more often at S8 than S7 (WB: fsm 0.17 to 0.67 on 24 instances, counter 0.00 to 0.39 on 18), and nothing else moves by 0.20 (Table LIX).

What this says for L Q3.2. The map already names, for every LUT, the set of statements it blends, so a detector could split a LUT region by that set instead of guessing; the mux fooling at abc and the compare loss at opt say where such a split is worth the most. This is a suggestion, not a result: no detector was changed in this round. Not done: the big designs, the ood split, a VPR flow, the instance axis across several instances of one module (units are keyed by construct, so two instances of one module are one unit), and the word level, which the map does not observe.

VI. Discussion

Three observations follow. First, the answer depends on what the flow erased. The structures that survived synthesis recover almost perfectly without any intelligence beyond reading the pass's own facts; the structures synthesis erased (the counter behind its flip-flop, the mux behind its gates) recover weakly, and no amount of honest Bayesian machinery changes that without better detectors or a model. Second, the graph features carry the recovery: the GNN only beats 1-WL when node, graphlet, WL and NPN features combine, which is the same lesson as the clustering table (community methods find the hierarchy, centroid methods do not). Third, the benchmarks made the limits visible: the renamed-instance protocol proved naming neutrality over 105,646 identifiers, and the same protocol exposed a rename-script defect (library .blackbox model names were renamed, which silently removed every hard proposal on renamed floors) and the spectral determinism defect – both fixed or reported, and both invisible without the protocol.

A fourth observation comes from round 2 (Section V-F): readable and verified are different properties, and the two tools hold one each. The language model produces

RTL whose similarity to the reference is high and whose operators it names (operator type-F1 0.465 on Q1 and Q2, correct samples elaborated to RTL), but nothing proves the text equals the netlist; Loki's output is formally equal to its input in all 150 runs and recovers few operators (0.081, calibrated f2 run; the f3 run recovers none), and its similarity to the reference is lower in every cell. The f3 failure of round 1 may be the detectors' vocabulary and not missing information (a LUT6 re-map reaches the f2 level in-sample but not on the held-out designs, 0.87 against 0.51, so the question stays open), the three refinements we expected to help (evidence-family weighting, the width of the must/may bracket as an uncertainty measure, a structural prefilter before the prover) did not on the registered tests, and the one new instrument that earned its place is the operator F1, which sees recovery where AST and CFG see only form.

A fifth observation comes from round 3 (Section V-G): many wrong candidates are pieces of real constructs, but the pieces do not tile the words. By the letter of the pre-registered rule about half of the false candidates are fragments of an instance of a related kind (H9a supported by the letter of the rule; not robust: 0.415 without one design, 0.304 without the near-tautological ff_en slices), and the noise class could not be filled because the ground truth covers every cell. A word's fragments rarely cover it (H9b not supported). Propagation from the I/O words reaches registers but not the arithmetic or the FSMs (H10a not supported; H10b is not supported but uninformative once H10a fails), and the FSM, decode, select chain covers 0.214 of the control muxes. The next detector is therefore word-first, not a regrouping of the present one.

A sixth observation comes from round 4 (Section V-H): this dictionary and this GNN, trained on synthetic idioms, did not transfer to real logic, and the flow destroys part of the style. The dictionary wins on no eligible kind (H12b not supported) and the GNN does not improve on real designs with more synthetic data (H12c not supported); on the twin set the style-recovery ceiling is 0.500 at f3.

A seventh observation comes from round 5 (Section V-I): a word-first detector built from structure raises the recall of enabled-register words from 0.008 to 0.146 at a precision of 0.014 (F1 0.026), and no hypothesis reaches its margin (H16a not supported, H16b inconclusive, H16c not supported, H16d inconclusive, H16e inconclusive, H16f not supported and degenerate). The word-level eq_const is a subset of the per-bit rule and adds nothing in the combined arm, the FSM test finds counters and registers as often as FSMs, and the tiling null is dominated by a few large candidates. Where the missed constructs went was not measured; saying which pass joined each of them to its neighbours is the loss audit (L15), which needs the Odin III provenance export.

An eighth observation comes from the comparison with HAL's DANA (Section V-J): on the evaluation sets the pre-registered rule returns inconclusive between the word-first detector and DANA (pooled F_J 0.169 against 0.261 at f2). DANA is a library-independent baseline for register words, and neither tool recovers most of them.

A ninth observation comes from round 6 (Section V-L): where Odin III loses a construct as a separable unit is mostly the first opt and abc, and the D14 join labels account for every new blend. Of 580 lost units, 320 are lost at opt and 166 at abc; the compare statements that round 5 expected to vanish into LUTs are already blended after opt, and abc fools the mux-word detector (all three arms).

VII. Conclusion

Loki III asks how far netlist-to-RTL reverse engineering gets without a language model. On the benchmarks of this paper the answer is: the already-reverse-engineered structures, perfectly; the rest, weakly but measurably, with calibration that reports its own error, a graph stage whose features beat 1-WL, and clusters that carry the designer's hierarchy. The methodology – provenance-tracked labelled datasets plus the renamed-instance protocol – is the part we would export to any future comparison, with or without a model.

Appendix A
Research log

This appendix records the order in which the questions were asked, what each result suggested, and what became of the next set of hypotheses, so that the progress of the work is visible and not only its end state. The decision rules of round 2 are in experiments/round2/EXPERIMENT.md; their thresholds were set by the implementer during the round (the file was committed after the scripts and the result CSVs), so they are not pre-registered. The rules of round 3 (experiments/round3/EXPERIMENT.md) were committed before any round-3 number was computed (commit 5f8de24).

Round 0 (2026-10-21 to 22): build. Question: how much structure does Loki recover without a language model? Built L0 (provenance-labelled, anonymised datasets and the renamed-instance neutrality check), L1 (Bayesian hypotheses with calibration and proofs) and L2 (graph analysis).

Round 1 (2026-10-23): first results. Section V-B to Section V-E. Hard blocks are fully recovered, because synthesis had already reverse engineered them; soft structures recover weakly; on the cmos6 gate netlist (f3) L1 recovers nothing and AST and CFG do not move; the GNN beats 1-WL; and the language model is strong on the simple VerilogEval quartiles but its output is unverified.

What round 1 suggested. (i) The f3 zero beside a weak f2 result: is the failure the detectors' LUT vocabulary, not missing information? (ii) The language-model and Loki figures sat on different benchmarks: put them on the same 150 designs. Our working position is that the model is

readable but unverified on small designs, Loki is verified but partial, and the combination is the contribution. (iii) The calibrated flow commits little: which evidence family earns that? (iv) The reverse-engineering result is a smaller graph bracketed by what is proven (must) and what is possible (may), in the style of the graph-sandwich problem: is the width of that bracket an uncertainty measure, and does a cheap structural test before the prover keep recall? (v) AST and CFG tie where the recovery differs: is there a metric that sees it?

Round 2 (2026-10-24): the hypotheses and what happened. Table LXI. The operator F1 is the one hypothesis the rule supports, by its letter and on weak evidence (H7). The LUT6 re-map reaches the f2 level in-sample (0.87) but not on the held-out split (0.51, too few matches for the rule), so H1 is inconclusive. The head-to-head puts Loki below the model on every cell with its output equal to its input in all runs, but the similarity does not stay flat with complexity (H2, inconclusive). The evidence families add nothing over the priors on held-out designs, because the calibration already zeroes or shrinks them (H4, inconclusive). The bracket width is not an uncertainty measure on the registered set (H5a, not supported; an exploratory look at calibrated runs that list candidates from $p = 0.1$ gives Spearman -0.346), and a structural prefilter cannot cut the prover without losing proved constructs (H5b, not supported). The head-to-head (Section V-F) fills in the positioning of the discussion.

What round 2 suggested. The calibrated flow commits almost nothing and the uncalibrated one proposes many wrong candidates. Are the wrong candidates fragments of real word constructs (a grouping problem) or unrelated (a detector problem)? Peter (2026-10-24) put the same question from the reverse-engineering side: "for Loki - the question is from a reverse engineering perspective why is it hard to reconstruct to HDL - going from if structures (which we know are muxes) and then down it seems like many of the patterns are constant - so hypothesizing that things are a bus (and by a bus I mean a signal that could be a traditional bus, but also a wide wordsize signal that has regular oprtations to it - for example figuing out that there is a word in the datapath or control then allows us to look for patterns to reconstruct. Control flow is usually in the form of muxin and FSMs have pattern of feedback (state equals next state) and then control based on state. How do we find these patterns better?" and "Also, the outputs (words) and the inputs as words also provide hints of where the words are".

Round 3 (2026-10-24): the hypotheses and what happened. Table LXII; Section V-G. Rules committed first (5f8de24). H9a (wrong candidates are related fragments) is supported by the letter of the rule, with related-fragment share 0.583 of 2,615 false candidates, but not robust: 0.415 without the largest design and 0.304 without ff_en (exploratory), and the noise class is unreachable by construction (the ground truth covers every cell); H9b (fragments cover the word) is not supported at 0.065; H10a (datapath words reached from the ports within two hops) is not supported at 0.095 and H10b (control is not) not supported at 0.605, uninformative by construction once H10a fails; the FSM chain (L11) is measured only. The next step is a word-first detector (register words by shared clock, enable and reset; datapath words by per-bit isomorphic cones; control nets by select fan-out; FSMs by feedback components and their decoders); L8 stays deferred.

Next. L3, a cross-suite GNN (train on one suite, test on the other); L6, a motif null model for the graphlet counts; and L8, a language model given the ReverseProvenance document (deferred by Peter), which is the experiment that tests the combination directly.

Round 4 (2026-10-25): the hypotheses and what happened. Table LXIII; Section V-H. Peter (2026-10-24): "Queue those as round 4 after round 3 finishes - we're trying to make these better - I like the Loki ideas so yes (especially the idea of patterns from our data (what about the bigger designs, do they provide other insights of

TABLE LXI
Research log: the round-2 questions, their status and the deciding number (all generated from experiments/round2).

RQ	Hypothesis	Round	Status	Deciding number
L1	f3 failure is detector vocabulary	2	inconclusive	micro-F1 LUT6/f2 = 0.87 (all, in-sample), 0.51 (test)
L2	Loki below the LLM on AST/CFG, flat over Q1–Q4	2	inconclusive	AST Q1/Q2 0.763/0.460 vs 0.894/0.761
L4	one or two evidence families carry the posterior	2	inconclusive	full minus priors-only micro-F1 −0.0013
L5a	bracket width falls with per-design F1	2	not supported	Spearman 0.087 (p = 0.417)
L5b	structural prefilter cuts prover calls, keeps recall	2	not supported	best zero-loss rule prunes 0.067
L7	operator F1 separates AST/CFG ties	2	supported by the letter of the rule (weak evidence)	tie pair dF1 0.0173; flatten dAST 0.0422
L3	cross-suite GNN	next	not run	–
L6	motif null model	next	not run	–
L8	LLM with reverse provenance	next	deferred (Peter)	–

TABLE LXII
Research log: the round-3 questions, their status and the deciding number (all generated from experiments/round3).

RQ	Hypothesis	Round	Status	Deciding number
L9a	wrong candidates are fragments of GT words	3	supported by the letter of the rule; not robust	related-fragment share 0.583 of 2,615 false; noise 0 by construction (every cell is in a GT word); without the largest design 0.415, without ff_en 0.304 (exploratory)
L9b	fragments of a word cover it	3	not supported	share with coverage $\geq$ 0.8: 0.065 (touched words only, exploratory: 0.069)
L10a	datapath words reached from I/O within 2 hops	3	not supported	share reached 0.095 (per-design mean 0.326, exploratory)
L10b	control words are not	3	not supported (uninformative by construction)	share reached 0.605 (per-design mean 0.286, exploratory)
L11	FSM chain: feedback, decode, select	3	measurement only	0.214 of muxes with a select are chain sinks

TABLE LXIII
Research log: the round-4 questions, their status under the Stage-0 rules and the deciding number (generated from experiments/round4).

RQ	Hypothesis	Round	Status	Deciding number
L12a	corpus is usable	4	supported	P_ok=1.0000 generated=1200 ok=1200 min_family_ok=100 ood=1.0000 train=1.0000
L12b	dictionary beats detectors (V0 as written)	4	not supported	won=0/7 floor=f2 sets=micro,vetest
L12b	same, V1 (added after Stage 0)	4	not supported	won=0/7 floor=f2 sets=micro,vetest
L12b	same, V2 (added after Stage 0)	4	not supported	won=0/7 floor=f2 sets=micro,vetest
L12b2	dictionary generalises over unseen regions (V0)	4	inconclusive	R=0.3288 iid=0.1866 ood=0.0614 (iid < 0.20: ratio uninformative)
L12c	GNN transfer improves with corpus size	4	not supported	slope=-0.0026/doubling CI=[-0.0097,0.0094] n_designs=29
L13a	big designs mostly repeated structure	4	supported	share_rep(big)=0.9283 (6653/7167) designs=6
L13b	big designs repeat more than small	4	supported	big=0.9283 small=0.6879 (6233/9061) diff=0.2404
L13c	repeated instances share the signature (V0)	4	supported	same_signature=0.8606 groups=911
L14a	flow destroys style information	4	supported	ceiling_f3=0.5000 f2=0.4823 f1=0.4697
L14b	coarser floor destroys more	4	inconclusive	f3-f2=0.0177
L14c	MDL ranks references above recovered RTL	4	inconclusive	l0-f2=0.860 l0-f3=0.913 f2-none=0.940 f3-none=0.940 f2-calib=0.867 f3-calib=0.913
L14d	MDL tracks round-2 similarity	4	inconclusive	rho=-0.5110 (p=1.1e-35) partial=0.3029 (p=2.1e-12) n=516

patterns).” What round 3 suggested: the wrong candidates are real constructs that do not tile the words, so a word-first detector needs labelled data and pattern knowledge. Rules committed first (c6c66bf); variants V1 and V2 of the dictionary were added after Stage 0 and before any dictionary result (entry 2026-10-25-a) and change no status. RQs: L12a corpus (H12a supported), L12b dictionary (H12b not supported, H12b2 inconclusive), L12c GNN scaling (H12c not supported), L13 big designs (H13a–c supported, supported, supported), L14 recoverability (H14a–d supported, inconclusive, inconclusive, inconclusive). Next: the word-first detector of the discussion, built and tested on the corpus. Statuses are those of the Stage-0 rules; exploratory checks (entry 2026-10-25-c) show H13b flipping with the small baseline (inconclusive against micros and VerilogEval, not supported against VerilogEval alone) and the repeat key near its null on small real designs, so H13b and the “free evidence” reading are uninformative or fragile by the letter of the rule; H13a and H13c stay informative.

Round 5 (2026-10-25): the hypotheses and what happened. Table LXIV; Section V-I. Question L Q2.4 (hub purpose page): can words, control nets and FSMs be found first, and the rest built around them? What round 3 and round 4 suggested: build words directly from structure, use the corpus as a test bed for grouping rules, use repetition only where it is real. Rules committed first (ce02fa4); a dated entry in experiments/round5/EXPERIMENT.md (2026-10-25-a) records the implementation choices, among them the relaxed FSM rule, disclosed late (the code of 6e97d88 already carried it); entries 2026-10-25-b and -c (review fixes) correct two statements of entry a (the FSM rule and the null seeding were deviations from the committed text), label H16f degenerate, correct the H16a reachability arithmetic and add exploratory measurements. No status changed. RQs: L16a tiling (H16a not supported), L16b enabled words (H16b inconclusive), L16c word-level eq_const (H16c not supported), L16d chains (H16d inconclusive), L16e growth against a null (H16e inconclusive), L16f repetition on big designs (H16f not supported, degenerate). Word grouping raises ff_en recall to 0.146 at a precision of 0.014 and nothing else reaches its margin; why is not measured here, and the provenance-export audit (L15) can attribute missing constructs to a pass.

TABLE LXIV
Round 5 (2026-10-25): the hypotheses and their status.

Hypothesis	Status
H16a	not supported
H16b	inconclusive
H16c	not supported
H16d	inconclusive
H16e	inconclusive
H16f	not supported (degenerate)

Round 6 (2026-10-27): the loss audit. Section V-L. Question L Q3.1 (hub purpose page): at which Odin III step is each construct first lost as a separable unit, joined with what, and why? What round 5 suggested: register and enable words are findable, decoders, compares and mux boundaries are absorbed into LUTs at f2. Rules committed first (6d0e5b5); four dated entries in experiments/round6/EXPERIMENT.md (two on development designs, two post-data and exploratory). RQs: L15a losses per kind and pass against a random-set null, L15b partners and reason classes, L15c the explained share (0.606, partly explained), L15d detector recall at S7 and S8 (only the mux word is fooled by abc). Outcome: 580 of 1436 units are lost, at opt (320), abc (166), flatten (68, instance cells) and the second opt (26), none at fsm, wreduce or lower; every unlabelled merge is an inherited blend. Compare statements are lost at opt, before any LUT exists.

References

[1] M. O. Faruque, P. Jamieson, A. Patooghy, and A.-H. A. Badawy, “Gate-breaker: An LLM-powered netlist-to-RTL reverse engineering tool,” in Proc. IEEE Int. Performance, Computing, and Communications Conf. (IPCCC), 2025, pp. 363–370.

[2] M. Liu, N. Pinckney, B. Khailany, and H. Ren, “VerilogEval: Evaluating large language models for Verilog code generation,” in Proc. IEEE/ACM Int. Conf. Computer-Aided Design (ICCAD), 2023.

[3] F. Gharibian, L. Shannon, and P. Jamieson, “Finding system-level information and analyzing its correlation to FPGA placement,” in Proc. Int. Conf. Field Programmable Logic and Applications (FPL), 2010, pp. 544–549.

[4] Embedded Security Group, “HAL - the hardware analyzer,” https://github.com/emsec/hal, 2019, mIT licence. Release v4.5.0 (2025-09-23).

[5] M. Fyrbiak, S. Wallat, P. Swierczynski, M. Hoffmann, S. Hoppach, M. Wilhelm, T. Weidlich, R. Tessier, and C. Paar, "HAL – the missing piece of the puzzle for hardware reverse engineering, trojan detection and insertion," IEEE Transactions on Dependable and Secure Computing, vol. 16, no. 3, pp. 498–510, 2019.

[6] N. Albartus, M. Hoffmann, S. Temme, L. Azriel, and C. Paar, "DANA: Universal dataflow analysis for gate-level netlist reverse engineering," IACR Transactions on Cryptographic Hardware and Embedded Systems (TCHES), 2020.

[7] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in Int. Conf. Learning Representations (ICLR), 2019.

[8] L. Alrahis, A. Sengupta, J. Knechtel, S. Patnaik, H. Saleh, B. Mohammad, M. Al-Qutayri, and O. Sinanoglu, "GNN-RE: Graph neural networks for reverse engineering of gate-level netlists," IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems, 2022.

[9] W. Li, A. Gascon, P. Subramanyan, W. Y. Tan, A. Tiwari, S. Malik, N. Shankar, and S. A. Seshia, "WordRev: Finding word-level structures in a sea of bit-level gates," in 2013 IEEE International Symposium on Hardware-Oriented Security and Trust (HOST), Austin, TX, USA, Jun. 2013, pp. 67–74.

[10] P. Jamieson, K. B. Kent, F. Gharibian, and L. Shannon, "Odin II - An Open-source Verilog HDL Synthesis tool for CAD Research," in Proceedings of the IEEE Symposium on Field-Programmable Custom Computing Machines, 2010, pp. 149–156. [Online]. Available: http://www.computer.org/portal/web/csdl/doi/10.1109/FCCM.2010.31

[11] A. Mishchenko, S. Chatterjee, and R. K. Brayton, "Improvements to technology mapping for LUT-based FPGAs," IEEE Transactions on CAD, vol. 26, no. 2, pp. 240–253, 2007. [Online]. Available: http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.68.9003